

ADDIS ABABA UNIVERSITY

College of Technology and Built Environment

School of Electrical and Computer Engineering

Stream: Computer Engineering

**Beyond Collateral: An Explainable AI
Framework for Alternative Data-Driven Credit
Scoring of Ethiopian SMEs**

A Bachelor of Engineering Project Submitted in Partial Fulfilment of the
Requirements for the Degree of Bachelor of Science in Computer
Engineering

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Declaration

We, Leykun Birhanu (UGR/7741/14) and Fikir Anteneh (UGR/0415/14), declare that this thesis entitled *Beyond Collateral: An Explainable AI Framework for Alternative Data-Driven Credit Scoring of Ethiopian SMEs* is our own original work. It has not been submitted for any degree or examination at any other university. All sources of information used have been duly acknowledged in the text.

Where we have drawn on others' work, this has been appropriately cited.

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Certificate of Approval

This is to certify that the project work entitled *Beyond Collateral: An Explainable AI Framework for Alternative Data-Driven Credit Scoring of Ethiopian SMEs* has been carried out by Leykun Birhanu (UGR/7741/14) and Fikir Anteneh (UGR/0415/14) under my supervision and guidance. The work is submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Engineering at the School of Electrical and Computer Engineering, Addis Ababa University.

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Abstract

Ethiopian Small and Medium Enterprises (SMEs) face a structural financing gap, trapped between microfinance institutions and commercial banks in a zone commonly referred to as the “Missing Middle.” Businesses requiring between 50,000 and 5,000,000 Ethiopian Birr (ETB) in working capital are systematically excluded from formal credit channels due to lack of collateral, thin credit files, and informal cash operations that leave no auditable financial trail. This thesis presents an Explainable Artificial Intelligence (XAI) framework for alternative data-driven SME credit scoring, developed in formal collaboration with the Commercial Bank of Ethiopia (CBE).

The system ingests 731 days of CBE transaction records for 100 businesses (approximately 73,100 daily heartbeat records) alongside matched loan histories and a 30-question web-based psychometric assessment capturing conscientiousness, integrity, financial risk tolerance, and delayed gratification as non-collateral character signals. A discounted cash flow NPV engine converts conservative P10 forecast outputs from a DeepAR probabilistic forecaster into defensible credit limit recommendations, complemented by XGBoost credit risk classification with SHAP explainability and an LSTM sequence encoder.

Experimental validation demonstrates that DeepAR achieves 2.94% Mean Absolute Percentage Error (MAPE) at a 30-day horizon with 94.20% empirical P10 coverage, substantially outperforming naive (85.76% MAPE) and SARIMA (77.39% MAPE) baselines. The full XGBoost credit risk model achieves AUC-ROC of 0.8842 and F1-score of 0.8140, representing a 24.21 AUC-point improvement over cash-flow features alone. The AI inference service is publicly deployed as a Dockerised FastAPI microservice on Hugging Face Spaces, enabling community inspection and future integration by licensed microfinance institutions.

Keywords: credit scoring, explainable AI, SME finance, cash-flow-based lending, DeepAR, XGBoost, SHAP, psychometrics, Ethiopia, microfinance, NPV.

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We also acknowledge the open-source communities behind XGBoost [1], SHAP [2], PyTorch Forecasting, GluonTS [3], FastAPI, and Laravel, whose tools form the technical foundation of this work.

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List of Abbreviations and Acronyms

AAU	Addis Ababa University
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
AUC-PR	Area Under the Precision-Recall Curve
CBE	Commercial Bank of Ethiopia
CFBL	Cash Flow-Based Lending
DDD	Domain-Driven Design
DeepAR	Deep Autoregressive Model
DFI	Development Finance Institution
DSCR	Debt Service Coverage Ratio
DTI	Debt-to-Income ratio
EFL	Entrepreneurial Finance Lab
ETB	Ethiopian Birr
F1	F1-score (harmonic mean of precision and recall)
FastAPI	Fast Application Programming Interface (Python framework)
GDP	Gross Domestic Product
HF	Hugging Face
HTTP	Hypertext Transfer Protocol
JWT	JSON Web Token
KDE	Kernel Density Estimate
KS	Kolmogorov-Smirnov statistic

LSTM	Long Short-Term Memory network
LTV	Loan-to-Value ratio
MAPE	Mean Absolute Percentage Error
MAE	Mean Absolute Error
MCC	Merchant Category Code
MFI	Microfinance Institution
ML	Machine Learning
NBE	National Bank of Ethiopia
NPV	Net Present Value
ORM	Object-Relational Mapping
PDP	Partial Dependence Plot
PHP	Hypertext Preprocessor
P10	10th percentile probabilistic forecast
P50	50th percentile probabilistic forecast (median)
P90	90th percentile probabilistic forecast
REST	Representational State Transfer
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
SARIMA	Seasonal Autoregressive Integrated Moving Average
SHAP	SHapley Additive exPlanations
SME	Small and Medium Enterprise
SMAPE	Symmetric Mean Absolute Percentage Error
SQL	Structured Query Language
UUID	Universally Unique Identifier
XAI	Explainable Artificial Intelligence
XGBoost	eXtreme Gradient Boosting

1.0 Executive Summary

Ethiopia’s Small and Medium Enterprise sector faces a structural credit exclusion problem that cannot be solved with conventional banking tools. Businesses seeking between ETB 50,000 and ETB 5,000,000 in working capital represent the productive backbone of the national economy, yet they fall into a financing gap between microfinance institutions and commercial banks. They are too large for MFI micro-loans and too informal for commercial bank collateral requirements. The result is a “Missing Middle” that suppresses economic growth and perpetuates inequality of opportunity.

This thesis presents a complete engineering solution: the **Ethio-SME Valuation System**, an Explainable AI framework that replaces collateral-based credit assessment with a multi-modal analysis of daily cash flow behaviour, credit repayment history, and psychometric character signals. The system was designed, built, tested, and deployed as a publicly accessible microservice within the scope of a Bachelor of Engineering final-year project at Addis Ababa University.

1.1 The Problem

Access to credit in Ethiopia is heavily collateral-constrained. The National Bank of Ethiopia’s regulatory framework, while modernising, still requires physical asset pledges for most SME loan products. For a business operating from a rented stall, processing daily cash transactions through a mobile platform, with no immovable property to pledge, there is no pathway into formal credit regardless of how reliably the owner manages cash. The system is structurally exclusionary, not merely imperfect.

The secondary barrier is information asymmetry. Even when a lender wishes to extend credit, they have no reliable mechanism to assess ability or willingness to repay for a business with a thin or absent formal financial history.

1.2 The Solution Architecture

The Ethio-SME Valuation System resolves both barriers through a three-model AI architecture that reads the financial behaviour already embedded in a business’s CBE transaction account:

- **DeepAR Probabilistic Forecaster:** A global deep autoregressive model trained across all 100 CBE businesses simultaneously produces P10, P50, and P90 cash flow forecasts. The P10 conservative lower bound serves as the credit-safe cash flow floor for capital sizing.
- **XGBoost Credit Risk Classifier:** A gradient-boosted ensemble classifier with

monotonic domain constraints and SHAP (SHapley Additive exPlanations) explainability. The model ingests 24 engineered features spanning transaction behaviour, loan repayment history, psychometric character scores, and macroeconomic indicators to produce a calibrated default probability.

- **LSTM Sequence Encoder:** A 2-layer Long Short-Term Memory network that encodes 60-day cash flow sequences into three named latent signals — risk, volatility, and trend — which are injected as additional features into the XGBoost scoring pipeline.

A fourth component, the NPV Credit Limit Engine, applies discounted cash flow principles to translate P10 forecast outputs into a legally defensible credit limit recommendation, automatically adjusting for CBE policy rate changes and the individual borrower’s psychometric composite score.

1.3 Data Foundation

The system was validated on a real institutional dataset obtained through a formal visit to the CBE headquarters in Addis Ababa. The Manager of the Information Management Department agreed to provide an anonymised research dataset of 100 SMEs with 731 consecutive days of daily financial heartbeats (approximately 73,100 records) and matched loan histories. This makes the system the first known multi-modal SME credit scoring framework in Ethiopia validated on real institutional transaction data.

1.4 Key Results

Table 1.1: Summary of Key Model Performance Metrics

Model	Metric	Value	Benchmark
DeepAR	MAPE (30-day horizon)	2.94%	Naive: 85.76%
DeepAR	P10 empirical coverage	94.20%	Target: >85%
XGBoost (full model)	AUC-ROC	0.8842	Target: >0.75
XGBoost (full model)	F1-Score	0.8140	—
XGBoost (cashflow only)	AUC-ROC	0.6421	Baseline
NPV Engine	Max credit capacity (365d)	ETB 2,411,450	—

The full XGBoost model incorporating psychometric scores and LSTM signals delivers a 24.21 AUC-point lift over cash-flow features alone, proving the incremental value of multi-modal data fusion.

1.5 Deployment

The AI inference service is deployed as a Dockerised FastAPI microservice on **Hugging Face Spaces** at <https://leykun-code-ethiopian-sme-ai-service.hf.space>. This public deployment enables any licensed microfinance institution or researcher to inspect the model card, interact with the API, and contribute improvements — positioning the framework as a foundation for a “Credit Scoring as a Service” integration layer for the Ethiopian MFI ecosystem.

2.0 Introduction

2.1 Background

Small and Medium Enterprises (SMEs) are widely recognised as the engine of employment and economic growth in developing economies. In sub-Saharan Africa, SMEs account for approximately 80% of employment and a significant share of GDP, yet they remain chronically underserved by formal financial systems [4]. Ethiopia presents a particularly acute version of this structural challenge.

The Ethiopian economy is characterised by a large informal sector, a banking system that remains heavily collateral-dependent, and an SME population whose financial transactions occur largely in cash or through informal channels. The National Bank of Ethiopia’s (NBE) regulatory landscape, though modernising with a suite of digital finance directives issued between 2020 and 2025, still presupposes a formal financial history as the basis for creditworthiness assessment [5].

The consequence is a phenomenon well-documented in development finance literature: the **“Missing Middle”** [4]. This refers to the population of businesses that require working capital in the range of ETB 50,000 to ETB 5,000,000 — too large for microfinance institution (MFI) micro-loans, which typically cap at ETB 30,000–50,000, and too informal and undocumented for commercial bank products that require collateral, audited financial statements, and a formal credit history. These businesses exist in a financing vacuum.

Ethiopia’s leading commercial banks, including the Commercial Bank of Ethiopia (CBE) — the country’s largest bank with approximately 40 million customers and over 1,700 branches — have the reach to serve this segment, but lack the credit assessment infrastructure to do so safely without collateral [5]. Meanwhile, the rapid expansion of digital banking through CBE’s Birr mobile platform and the emergence of fintech players such as Chapa have created unprecedented digital transaction trails for a segment that was previously financially invisible.

2.2 Engineering Context and Motivation

The initial framing of this research, as documented in the project proposal, was a conventional binary default prediction problem: train a classifier on historical default/no-default labels and use it to predict future default probability. This approach is well-established in the credit risk literature and underpins most commercially deployed credit scoring systems.

However, early engagement with the realities of the Ethiopian SME data landscape

exposed three fundamental barriers that render this approach inadequate in this context:

1. **The Data Deadlock:** Binary default classification requires a substantial volume of labeled default events. In the CBE dataset, the distribution of repayment outcomes is heavily imbalanced — the majority of the 100 businesses in the dataset maintained performing loan status. Training a default classifier on a dataset with very few positive (default) labels yields a model that is nominally accurate but practically useless, tending to predict “no default” for every input.
2. **The Thin-File Exclusion Trap:** Any model trained on labeled defaults perpetuates the problem it is meant to solve. The businesses with the richest default labels are those who were historically extended credit — which is precisely the population that was not excluded by collateral requirements. The uncollateralised “Missing Middle” has no formal loan history at all, so a default-based model cannot score them. It would simply reproduce the existing system’s exclusions.
3. **The Backward-Looking Bias:** Default prediction optimises for past behaviour patterns under a specific economic regime (e.g., 2020–2022 Ethiopian economic conditions). Cash flow dynamics in Ethiopia shifted significantly in this period due to inflation, regional conflict, and foreign exchange pressures. A model calibrated on past defaults is a poor guide to the future creditworthiness of forward-looking, adaptive businesses.

These barriers motivated a strategic pivot, documented in the project’s Phase III revision, from binary default prediction to **Cash Flow-Based Valuation (CFBL)** [6]. Rather than asking “will this business default?” the question becomes “what is the reliable cash-generating capacity of this business, and how much credit can it safely service?” This reframing is consistent with the most advanced international SME lending practice [4] and is the methodology advocated for thin-file markets by the World Bank and FSD Kenya.

2.3 Problem Statement

There is no established, publicly available, and practically deployable credit scoring framework for Ethiopian SMEs that:

1. Relies on alternative data (daily transaction behaviour, psychometric character signals) rather than collateral or formal financial statements;
2. Produces both a risk score *and* a defensible cash-flow-based credit limit recommendation;
3. Provides regulatorily compliant explainability (adverse action reason codes, SHAP feature attributions);

4. Is validated on real Ethiopian institutional transaction data; and
5. Is publicly deployable as a microservice that licensed MFIs can integrate.

The absence of such a framework perpetuates the collateral trap and leaves a substantial portion of Ethiopia's economic potential unfinanced.

2.4 Objectives

2.4.1 General Objective

To build and validate an Explainable AI (XAI) framework for alternative data-driven SME credit scoring in the Ethiopian context, using real CBE transaction data, psychometric character signals, and interpretable machine learning models.

2.4.2 Specific Objectives

1. **SO1:** Develop a multi-model AI inference service incorporating XGBoost credit risk classification, DeepAR probabilistic cash flow forecasting, and an LSTM sequence encoder.
2. **SO2:** Engineer a robust CBE transaction feature pipeline that transforms raw daily heartbeat records into a structured 24-feature vector capturing cash flow dynamics, volatility, trend, and seasonality.
3. **SO3:** Integrate a web-based psychometric assessment module (conscientiousness, integrity, financial risk tolerance, and delayed gratification) into the credit scoring pipeline as a non-collateral character signal.
4. **SO4:** Build an NPV-based credit limit engine that translates probabilistic P10 cash flow forecasts into conservative, policy-rate-adjusted credit limit recommendations.
5. **SO5:** Deploy the AI inference service as a publicly accessible Dockerised FastAPI microservice on Hugging Face Spaces, enabling open community inspection and future third-party integration.
6. **SO6:** Validate all three model components on the CBE dataset using rigorous held-out test evaluation, ablation studies, and baseline comparisons.
7. **SO7:** Implement a live data ingestion microservice connecting directly to partner bank APIs to fetch real-time transaction records. *This objective was descoped after CBE provided the dataset as structured .xlsx files; direct API connectivity was found to be legally prohibited under NBE data-sharing regulations and CBE's core banking security policy. Manual upload mechanisms were used instead.*

2.5 Scope of the Project

This project is scoped as a Bachelor of Engineering proof-of-concept system. The following are explicitly within scope:

- Design, development, and testing of the AI inference microservice and the Laravel-based Core Application Service (loan workflow, psychometric portal, lender dashboard).
- Training and validation of DeepAR, XGBoost, and LSTM models on the CBE dataset, including the full PostgreSQL database schema design.
- Public deployment of the AI service on Hugging Face Spaces.

The following are explicitly outside scope:

- Live integration with CBE's production banking systems or real-time NBE macroeconomic data scraping (static import is used in both cases).
- Regulatory approval for commercial deployment under NBE Directive No. FIS/01/2024.
- A live pilot loan disbursement; the system produces AI-assisted recommendations requiring human lender review.

2.6 Report Structure

Chapter 3 surveys SME credit scoring theory, CFBL evidence, machine learning in credit risk, XAI, psychometrics, and the Ethiopian financial context. Chapter 4 specifies functional and non-functional requirements and success criteria. Chapter 5 presents the system architecture, database schema, and design trade-offs. Chapter 6 describes the data sources, feature pipeline, model architectures, and valuation framework. Chapter 7 details the implementation of each component. Chapter 8 documents the testing strategy and validation results. Chapter 9 presents quantitative results, market insights, and failure analysis. Chapter 10 maps outcomes to objectives, discusses limitations, and recommends future work.

3.0 Literature Review

3.1 SME Credit Scoring: Traditional Approaches and Their Limitations

Credit scoring is the systematic quantification of the probability that a borrower will repay a debt obligation as agreed [4]. Classical credit scoring models — the dominant approach in commercial banking globally — rely on structured financial data such as balance sheet ratios, income statements, credit bureau records, and collateral valuation to assess creditworthiness. The World Bank’s Credit Scoring Approaches Guidelines [4] provides a comprehensive taxonomy of these methods, classifying them along two axes: the type of data used (transactional vs. sociodemographic vs. psychographic) and the modelling technique (statistical vs. judgemental vs. machine learning).

In emerging market economies, and in sub-Saharan Africa specifically, these approaches share a common precondition — a sufficiently large population of borrowers with formal credit history — that fails for a substantial fraction of the SME population. Beck et al. [7] demonstrate that in low-income countries, only 15–25% of firms have access to a bank loan or line of credit, compared to 60–80% in high-income countries. The primary constraint cited by SMEs in these markets is not interest rate cost but collateral requirements — a finding consistent with World Bank Enterprise Survey data across 150 countries.

The collateral gap is especially severe for microentrepreneurs and small businesses in informal economies. Collateral-based lending produces a self-reinforcing exclusion cycle: businesses without a lending history cannot obtain credit without collateral, and without credit access they cannot build the capital base or financial history that would eventually satisfy collateral requirements. This is the structural origin of the “Missing Middle” financing gap.

3.2 Cash Flow-Based Lending: Theory and Evidence

Cash Flow-Based Lending (CFBL) represents a fundamental departure from collateral-dependent credit assessment. Rather than asking what assets the borrower can pledge, CFBL asks what cash flows the borrower reliably generates and how much debt those cash flows can service [6].

The emergence of mobile money and digital banking in Africa has changed the economics of cash flow measurement for small businesses. FSD Kenya’s technical note on Cash Flow Based Lending for SMEs [6] documents that mobile money transaction records — even relatively simple daily flow data — contain sufficient information to characterise an SME’s revenue trajectory, volatility, seasonality, and trend. The Kenya

study demonstrates that lenders using cash flow-based models achieve comparable or superior credit loss rates to collateral-based approaches, while extending credit to a substantially larger and more economically diverse population.

Subsequent IFC and McKinsey Global Institute analyses [8, 9] reach consistent conclusions: digital transaction data is the most reliable alternative signal for informal SME credit risk in markets without formal records.

For Ethiopia specifically, the rapid growth of CBE’s Birr mobile banking platform — from fewer than 5 million users in 2019 to over 40 million users by 2024 [5] — has generated precisely the kind of institutional daily transaction records that CFBL methodologies require. This project is the first known academic implementation of CFBL principles using real CBE SME transaction data.

3.3 Machine Learning in Credit Risk

3.3.1 Gradient Boosting and XGBoost

Gradient boosted decision trees have emerged as the dominant machine learning technique for structured-data credit risk problems. Chen and Guestrin’s XGBoost algorithm [1], published at KDD 2016, represents the state-of-the-art in this class, combining a regularised objective function with efficient tree construction and support for monotonic constraints.

The XGBoost training objective at boosting round t is:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t) \quad (3.1)$$

where l is a differentiable loss function, $\hat{y}_i^{(t-1)}$ is the prediction from the previous iteration, f_t is the new tree being added, and $\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ is a regularisation term penalising tree complexity T and leaf weight magnitude w .

For credit applications, the monotonic constraint feature of XGBoost is particularly valuable [1]. In credit risk, domain experts can specify that certain feature-to-risk relationships must be logically monotonic: higher missed payment ratios must never decrease the model’s default probability estimate, and higher net cash flow must never increase it. These constraints prevent the model from learning spurious correlations that violate financial logic. Gupta et al. [10] demonstrate that monotonically constrained XGBoost models are more stable, more interpretable, and produce fewer anomalous credit decisions.

Empirical evidence across multiple credit scoring benchmarks confirms XGBoost’s superiority over logistic regression and other ensemble methods for credit default prediction tasks [11]. On the German Credit dataset, the Home Credit Default Risk competition, and the Taiwan credit card dataset, XGBoost consistently achieves the highest AUC-ROC

among classical ML methods.

3.3.2 LSTM Networks for Financial Time Series

Long Short-Term Memory (LSTM) networks, introduced by Hochreiter and Schmidhuber [12], address the vanishing gradient problem that limits standard recurrent neural networks in capturing long-range temporal dependencies. An LSTM cell maintains a gated cell state c_t , updating it via input gate i_t , forget gate f_t , and output gate o_t :

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (3.2)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (3.3)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (3.4)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (3.5)$$

$$h_t = o_t \odot \tanh(c_t) \quad (3.6)$$

In financial time series applications, LSTMs have been applied to stock price prediction [13], macroeconomic forecasting, and credit risk assessment [14]. Their ability to capture non-linear temporal patterns over multi-month windows makes them well-suited to encoding the seasonal and cyclical patterns in SME cash flows.

In the present system, the LSTM is not used as a direct predictor but as a sequence encoder: it compresses a 60-day window of cash flow, inflow, and transaction count into three interpretable latent signals (risk, volatility, trend) that are injected as additional features into the XGBoost model. This hybrid architecture preserves XGBoost’s tabular performance advantages while incorporating deep temporal representations.

3.3.3 Probabilistic Forecasting with DeepAR

DeepAR, introduced by Salinas et al. [15] at Amazon Research, is a global autoregressive probabilistic forecasting model that learns shared patterns across multiple related time series simultaneously. Unlike per-series models such as ARIMA or Exponential Smoothing, DeepAR trains a single recurrent neural network across all series in the training set, enabling cross-series knowledge transfer — particularly valuable when individual series are short or noisy.

The DeepAR model parameterises a likelihood function $p(z_{i,t} | \theta(h_{i,t}))$ over the target variable, where $h_{i,t}$ is the hidden state of the LSTM encoder for series i at time t , and θ maps hidden states to distribution parameters. The training objective is the negative log-likelihood:

$$\mathcal{L} = - \sum_{i=1}^N \sum_{t=t_0}^T \log p(z_{i,t} | \theta(h_{i,t})) \quad (3.7)$$

At inference time, DeepAR generates Monte Carlo samples from the learned distribution, from which arbitrary quantiles (P10, P50, P90) are derived. The P10 quantile

is particularly important for credit applications: it represents the conservative 10th percentile forecast, below which only 10% of cash flow realisations are expected to fall. Using P10 as the credit sizing anchor provides a structural safety margin against downside risk.

The GluonTS library [3] provides the reference implementation of DeepAR and has been validated on the M4 forecasting competition dataset, M5, and numerous real-world demand forecasting benchmarks.

3.4 Explainable AI in Finance

The regulatory imperative for explainability in algorithmic credit decisions is well-established. The European Union’s General Data Protection Regulation (GDPR), Articles 13–15, creates a “right to explanation” for automated decisions that significantly affect individuals. The US Equal Credit Opportunity Act (ECOA) and Fair Credit Reporting Act (FCRA) require that adverse credit decisions be accompanied by specific reason codes communicating the most significant factors that influenced the decision. In Ethiopia, NBE’s digital finance directives implicitly require that automated credit decisions not be opaque or arbitrary [5].

Lundberg and Lee’s SHAP (SHapley Additive exPlanations) framework [2], presented at NeurIPS 2017, provides a theoretically grounded approach to explaining individual model predictions. SHAP is rooted in cooperative game theory: the Shapley value of feature j in prediction $\hat{y}_i$ is the average marginal contribution of feature j across all possible feature subsets. SHAP values satisfy three axioms (efficiency, symmetry, dummy) that make them the unique attribution method satisfying a natural set of desiderata.

For tree-based models, Lundberg et al. introduced TreeSHAP [16], an efficient $O(TLD^2)$ algorithm where T is the number of trees, L the maximum number of leaves, and D the maximum tree depth. TreeSHAP computes exact Shapley values orders of magnitude faster than model-agnostic approximations.

In credit applications, SHAP provides two critical outputs:

- **Global feature importance:** The mean absolute SHAP value across the test set ranks features by their aggregate influence on credit risk scores.
- **Local reason codes:** For an individual borrower, the top risk-increasing SHAP contributors provide human-readable adverse action reason codes compliant with regulatory requirements.

Recent work by Chen et al. [17] demonstrates that SHAP-based explanations can also serve as a foundation for fairness auditing by enabling comparison of SHAP value distributions across protected groups.

3.5 Psychometrics as a Credit Assessment Tool

The use of psychometric instruments in credit assessment — measuring personal character traits correlated with financial behaviour — was formalised by the Entrepreneurial Finance Lab (EFL) at Harvard Kennedy School and subsequently scaled by the World Bank’s Doing Business initiative. Arraiz et al. [18] evaluated the EFL psychometric credit assessment tool across 12 developing countries, including several in sub-Saharan Africa, and found that psychometric scores are statistically significant predictors of loan repayment behaviour, with the strongest effect sizes in markets where formal financial history is absent.

The key insight from this work is that for unbanked or thin-file borrowers, a well-designed psychometric assessment capturing integrity, conscientiousness, and financial risk attitudes adds predictive power that no amount of transaction data alone can provide. Character predicts repayment over and above financial capacity, because willingness to repay is distinct from ability to repay.

Klinger et al. [19] demonstrate in a Kenyan SME study that psychometric-augmented credit models reduce default rates by approximately 20% relative to credit-score-only models, and extend credit access to a 30% larger eligible population while maintaining credit loss targets. This evidence base supports the inclusion of psychometric signals as a first-class feature tier in the present system.

3.6 Ethiopian Financial Context

Four structural features of the Ethiopian financial landscape directly shape the feature engineering and modelling choices in this thesis [5]:

- **Informal economy:** The IMF estimates Ethiopia’s informal sector at 30–40% of GDP, so SME revenue captured in bank records frequently understates true economic activity.
- **Ethiopian calendar seasonality:** Major Ge’ez-calendar holidays — Genna (7 Jan), Fasika (Easter), Meskel (27 Sep), and Eid dates — create sector-specific cash flow spikes and troughs that Gregorian-trained models systematically miss.
- **Payday effect:** Public sector salaries and many private payrolls are disbursed in the 25th–30th window of each month, producing predictable retail inflow surges that must be captured as rolling-window features.
- **NBE policy rate:** Ethiopia’s reference rate rose from 7% (2023) to 10% (2024); the NPV credit limit formula is parameterised on this rate so that limits automatically tighten during monetary policy tightening cycles.

3.7 Research Gap

The literature surveyed in this chapter identifies the following research gap that this thesis addresses:

No prior published work combines multi-modal deep learning (LSTM sequence encoding + DeepAR probabilistic forecasting) with psychometric character signals and real Ethiopian institutional transaction data for SME credit scoring with full SHAP-based explainability and NPV-calibrated credit limit generation.

Existing approaches fall into one of the following limited categories:

- Binary default classification using mobile money aggregates (lacks explainability, fails for thin-file borrowers);
- Psychometric scoring in isolation (lacks cash flow validation);
- Probabilistic forecasting for demand planning (not credit-risk-adjusted);
- General-purpose credit ML applied to African data without Ethiopian- specific feature engineering.

This thesis provides the first integrated framework bridging all four components, validated on a real institutional dataset from Ethiopia's largest bank.

4.0 Requirements and Constraints

4.1 Functional Requirements

The functional requirements specify the observable behaviours the system must exhibit. Each requirement is identified by a unique code (FR1–FR6).

Table 4.1: Functional Requirements

ID	Name	Description
FR1	Data Ingestion	The system shall accept daily SME financial heartbeat records (inflow, outflow, transaction count, balance, channel) and persist them to the <code>sme_daily_heartbeat</code> table. Records shall be traceable to a data source type (<code>cbe_training</code> , <code>synthetic</code> , <code>app</code>).
FR2	AI Inference	Given a business UUID, the system shall produce a credit risk score (0–1), risk band (low/medium/high), P10/P50/P90 cash flow forecasts for a configurable horizon, SHAP-derived reason codes (up to 3), and an NPV credit limit recommendation (ETB).
FR3	Psychometric Assessment	The system shall present a 30-question web-based psychometric assessment to SME applicants. It shall compute normalised scores for four dimensions: integrity, conscientiousness, financial risk tolerance, and delayed gratification, and persist all responses and scores.
FR4	Credit Limit Calculation	The system shall compute the NPV credit limit using the P10 cash flow forecast, the NBE policy rate, the borrower’s risk band, and the composite psychometric score. The limit shall be bounded to [ETB 5,000, ETB 5,000,000].
FR5	Lender Dashboard	The system shall provide a loan officer dashboard displaying the list of loan applications, their AI valuation status, credit scores, SHAP reason codes, and forecast charts. The dashboard shall support approve/reject actions with mandatory reason code confirmation.

Continued on next page...

ID	Name	Description
FR6	Explainability Out-put	Every credit decision shall be accompanied by a SHAP waterfall explanation identifying the top risk-increasing features, their directions, and magnitude. The explanation shall be stored in the <code>shap_explanations</code> table for audit purposes.

4.2 Non-Functional Requirements

Table 4.2: Non-Functional Requirements

ID	Attribute	Requirement
NFR1	Latency	The <code>POST /predict</code> endpoint shall complete within 25 seconds (wall-clock) under normal database connectivity. Requests exceeding this threshold shall return HTTP 504 with a structured error.
NFR2	Forecast Accuracy	DeepAR shall achieve $\text{MAPE} \leq 30\%$ at the 30-day horizon. The P10 empirical coverage (fraction of actual values falling above P10) shall exceed 85%.
NFR3	Credit Risk Accuracy	The XGBoost model shall achieve $\text{AUC-ROC} \geq 0.75$ on the held-out test split. Monotonic constraints shall be enforced for all domain-constrained features.
NFR4	Scalability	The AI service Docker image shall be deployable on Hugging Face Spaces (2 vCPU / 16 GB RAM free tier) without modification.
NFR5	Security	All non-public API routes shall require a shared <code>X-Internal-Key</code> header. User authentication shall use JWT tokens with access and refresh token pairs. All passwords shall be bcrypt-hashed.
NFR6	Explainability	SHAP integrity check: $\text{sigmoid}(\text{expected_value} + \sum_j \text{SHAP}_j)$ shall equal the model output probability within a tolerance of 10^{-4} .
NFR7	Maintainability	Each model component shall be individually replaceable without affecting the other components. Model artifacts shall be versioned and stored under <code>data/models/<model_name>/</code> .

4.3 Engineering Constraints

1. **Data Privacy:** The CBE dataset was provided under an academic research agreement. All business identifiers must be anonymised using UUID hashing. Raw

transaction data may not be exposed via any public API. Only aggregated heartbeat records are used by the AI service.

2. **Regulatory (NBE):** The system produces credit scoring recommendations. It does not constitute a licensed credit decision system under NBE Directive No. FIS/01/2024 on AI in financial services. All outputs must be clearly labelled as AI-assisted recommendations requiring human lender review.
3. **Computational Resources:** Model training must be feasible on Google Colab (T4 GPU, 15 GB VRAM, 12-hour session limit) for portability and cost. Inference must run on the Hugging Face Spaces free tier (CPU-only, 16 GB RAM).
4. **Academic Timeline:** The project was executed in two semesters across five phases: (I) literature and proposal, (II) data acquisition and feature engineering, (III) model development, (IV) system integration and deployment, (V) testing and thesis writing. Phase III experienced a strategic pivot from classification to CFBL methodology, accommodated within the timeline.
5. **Open Source:** All system components must use open-source libraries and frameworks. No proprietary ML platforms (e.g., Azure ML, AWS SageMaker) shall be required for training or inference.

4.4 Success Criteria

The project is considered successful if all of the following criteria are met:

Table 4.3: Measurable Success Criteria

ID	Criterion	Target	Status
SC1	DeepAR MAPE (30-day)	$\leq 30\%$	✓ 2.94%
SC2	DeepAR P10 empirical coverage	$\geq 85\%$	✓ 94.20%
SC3	XGBoost AUC-ROC (test set)	≥ 0.75	✓ 0.8842
SC4	SHAP integrity check pass rate	100%	✓
SC5	System uptime on Hugging Face	Publicly accessible	✓
SC6	End-to-end prediction latency	≤ 25 s	✓
SC7	CBE dataset validated	100 businesses, 731 days	✓

5.0 System Design and Architecture

5.1 High-Level Design

The Ethio-SME Valuation System is designed as a loosely coupled microservice architecture comprising four independently deployable components. This separation of concerns isolates the AI inference workload — which requires Python, PyTorch, and XGBoost — from the web application workload, which uses the Laravel PHP framework and a React frontend. Both services share a single PostgreSQL database as the system-of-record, with a strict access contract: the Laravel Core Service holds write authority over all tables, while the AI Inference Service holds read-only access to training and inference data.

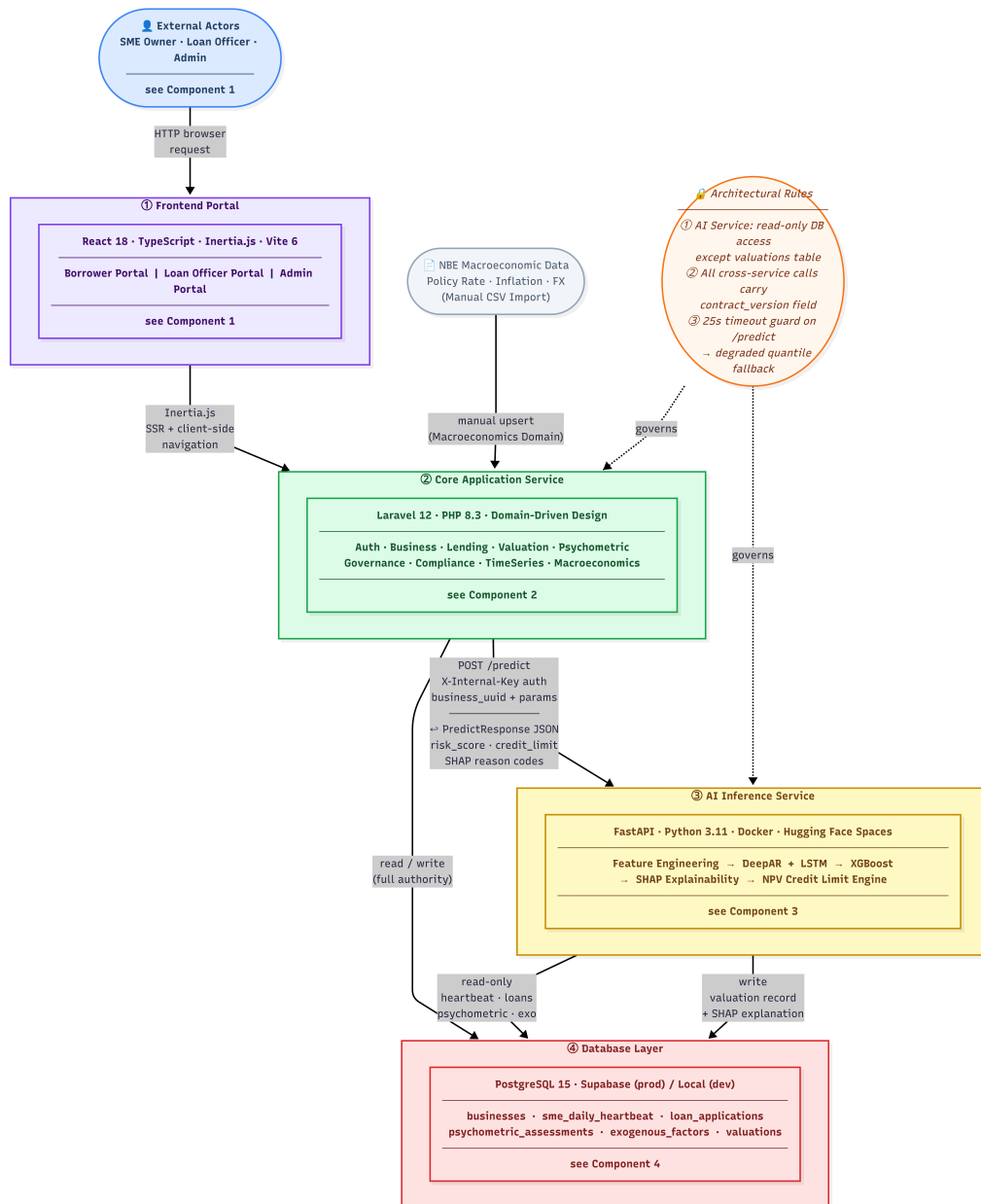


Figure 5.1: High-Level System Architecture of the Ethio-SME Valuation System

The four microservices and their responsibilities are:

1. **AI Inference Service (FastAPI, Python):** Self-contained FastAPI microservice responsible for model training, cash flow forecasting (DeepAR), credit risk scoring (XGBoost + LSTM), SHAP explainability, NPV credit limit computation, and drift monitoring. Deployed on Hugging Face Spaces as a Dockerised container.
2. **Core Application Service (Laravel 13, PHP):** Domain-Driven Design (DDD) backend API serving as the system's orchestration layer. Handles user authentication (JWT), business profile management, loan application state machine,

psychometric assessment submission, valuation orchestration (calling the AI service), and audit logging. Exposes a REST JSON API consumed by the frontend.

3. **Frontend Portal (React 18 / TypeScript / Inertia.js):** Three role-differentiated portals — SME Owner (borrower), Loan Officer (lender), and Super Admin — built on React with Tailwind CSS. Inertia.js bridges the frontend to Laravel’s routing without a separate API, enabling server-side rendering with client-side interactivity. Served through Vite 6.
4. **Database Layer (PostgreSQL / Supabase):** The persistent system-of-record for all identities, loan decisions, time-series aggregates, model predictions, psychometric responses, and audit logs. Hosted on Supabase for the production deployment, with local PostgreSQL for development.

5.1.1 Key Architectural Rules

- **Read-only AI service database access:** The FastAPI service never writes business or user data. It only reads pre-aggregated heartbeat records, loan history, psychometric scores, and exogenous factors. This boundary enforces NBE-aligned data minimisation.
- **Contract versioning:** All cross-service calls include a `contract_version` field. The Core Service stores the exact feature snapshot sent to inference, ensuring reproducibility of credit decisions for audit purposes.
- **Degraded mode:** If the AI service exceeds the 25-second timeout, Laravel returns a structured error object with a `retry_after_ms` field. For the thesis demo, the AI service includes a rolling quantile fallback forecaster that activates when the primary DeepAR model is unavailable.

5.2 Database Schema Design

The PostgreSQL schema is organised into six core tables, designed to capture the complete lifecycle of an SME’s relationship with the credit system.

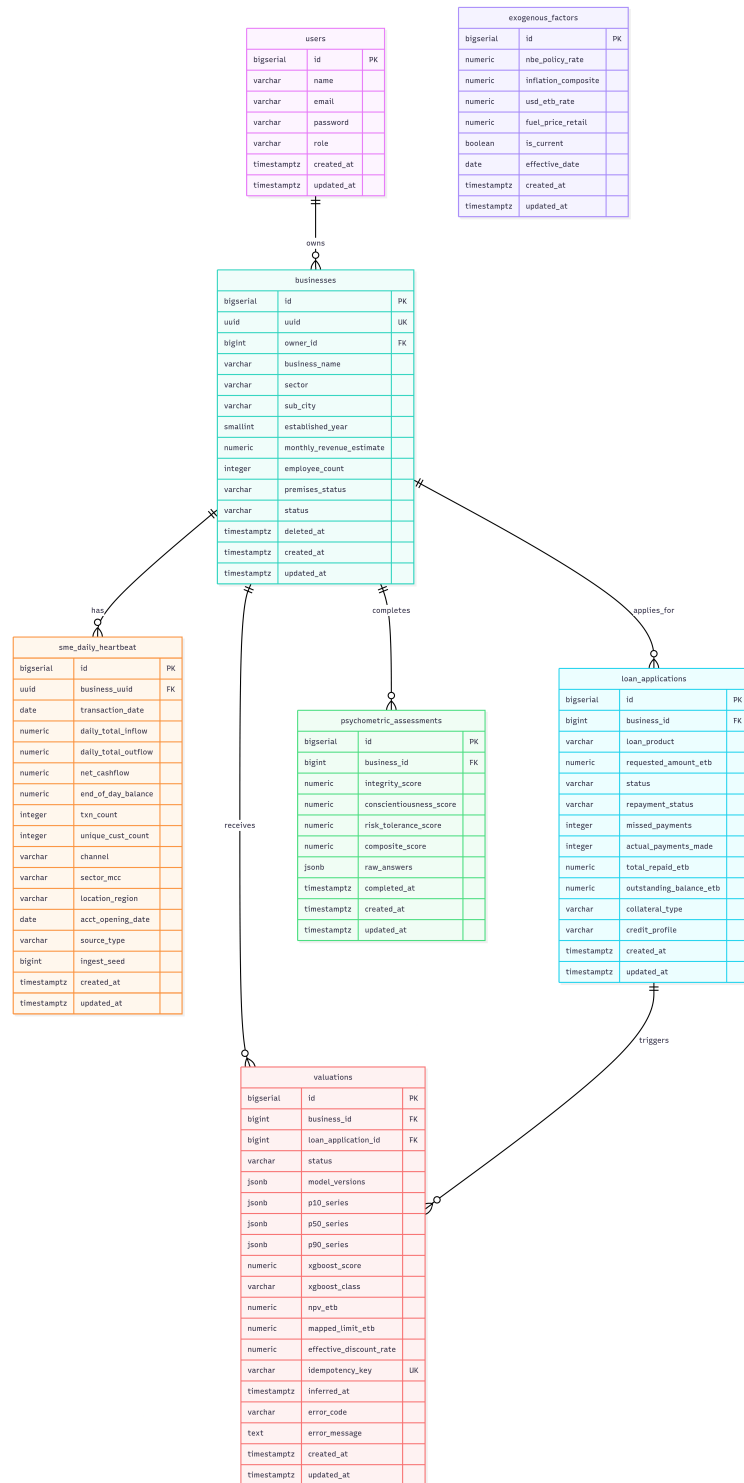


Figure 5.2: PostgreSQL Database Schema — Entity Relationship Diagram

5.2.1 Entity Relationship Diagram

The six core tables span the complete SME credit lifecycle: **businesses** is the central entity, referenced by **sme_daily_heartbeat** (daily transaction aggregates), **loan_applications** (loan lifecycle and repayment labels), **psychometric_assessments** (character scores), and **valuations** (AI inference outputs). The **exogenous_factors**

table holds the single active macroeconomic parameter row used by the NPV engine.

5.3 AI Service Architecture

The AI Inference Service is a self-contained Python microservice built with FastAPI. Its internal architecture follows a 15-step orchestration pipeline from raw business UUID to a structured credit valuation response.

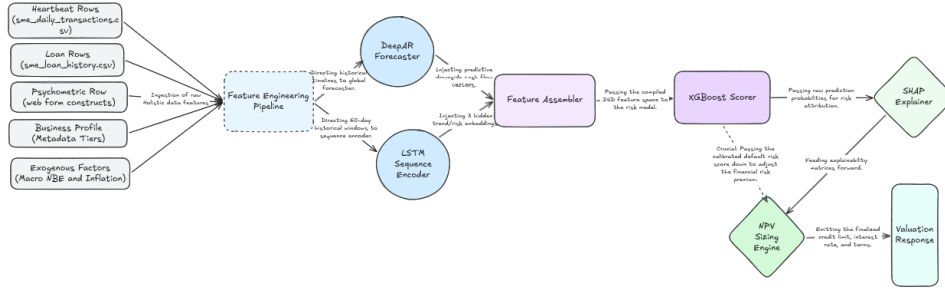


Figure 5.3: AI Inference Service Internal Architecture and Orchestration Pipeline

The 15-step prediction pipeline executes as follows:

1. Resolve the `businesses` row by UUID.
2. Load up to 365 days of `sme_daily_heartbeat` records.
3. Load loan history rows from the application tables.
4. Load the most recent psychometric assessment.
5. Fetch current `exogenous_factors` (NBE rate, inflation, USD/ETB).
6. Execute DeepAR to produce P10/P50/P90 cash flow forecast series.
7. Compute LSTM sequence encoding (60-day window), producing three named signals.
8. Assemble the 24–27 dimensional feature vector via the `FeatureAssembler`.
9. Run XGBoost inference to produce risk score, band, and raw SHAP values.
10. Map SHAP values to human-readable adverse action reason codes.
11. Compute NPV credit limit from P10 forecast using the discount rate formula.
12. Apply the minimum (ETB 5,000) and maximum (ETB 5,000,000) credit bounds.
13. Validate SHAP integrity ($\text{sigmoid}(\text{base} + \sum \text{SHAP}) \approx \text{risk_score}$).

14. Persist the valuation record, SHAP explanation, and drift metrics to PostgreSQL.
15. Return the structured `PredictResponse` JSON to the caller.

5.4 Design Choices and Trade-offs

5.4.1 *Microservices vs. Monolith*

A monolithic architecture combining the Python AI logic with the PHP/Laravel web application was considered during the initial design phase. This would have simplified deployment but would have introduced several unacceptable constraints: Python and PHP cannot share a process; the AI service requires PyTorch and XGBoost, adding approximately 2 GB of dependencies that would bloat the web server; and model retraining jobs would compete with HTTP request handling for CPU resources. The microservice split keeps each component independently scalable and replaceable.

5.4.2 *FastAPI vs. Flask*

FastAPI was selected over Flask for the AI service because of its native support for asynchronous request handling (critical for the 25-second prediction timeout), automatic OpenAPI schema generation (enabling Swagger UI for thesis demonstration), and Pydantic-based request/response validation that catches malformed inputs at the framework layer before they reach model inference code.

5.4.3 *PostgreSQL vs. Time-Series Database*

Time-series databases (InfluxDB, TimescaleDB) were evaluated as alternatives for the `sme_daily_heartbeat` table. However, the daily granularity of the heartbeat data — one row per business per day — does not create the extreme write throughput for which columnar time-series databases are optimised. Standard PostgreSQL with a composite B-tree index on `(business_uuid, transaction_date)` provides query performance well within the latency budget for the 730-day history window. The additional operational overhead of a specialist time-series engine was not justified for this data scale.

5.4.4 *Three-Model Ensemble vs. Single Model*

The three-model architecture was chosen because no single model can simultaneously serve all three required outputs: a *probabilistic cash flow forecast* (DeepAR), a *credit risk score with reason codes* (XGBoost + SHAP), and a *temporal sequence encoding* (LSTM). Attempting to collapse these into a single model would sacrifice either interpretability (a pure deep learning end-to-end model) or temporal depth (a tabular-only model). The ensemble approach assigns each model the task for which it is most architecturally suited.

6.0 Materials and Methods

6.1 Tools, Platforms, and Languages

Table 6.1: Technology Stack

Tool / Library	Version	Role in System
Python 3.11	3.11.x	AI Inference Service runtime
FastAPI	0.111+	AI service web framework; automatic OpenAPI docs
XGBoost	3.1+	Credit risk classification with SHAP
PyTorch	2.x	LSTM encoder training and inference
PyTorch Forecasting	—	DeepAR global probabilistic forecaster
GluonTS	—	Time-series utilities; DeepAR reference implementation
SHAP	0.50+	TreeExplainer for XGBoost; reason code generation
scikit-learn	1.4+	MinMaxScaler, train/test split
pandas / NumPy	—	Data wrangling, feature engineering
psycopg2	2.9+	PostgreSQL driver for Python
structlog	—	Structured JSON logging
Docker	—	Containerisation of AI service
Hugging Face Spaces	—	Public deployment (Docker SDK, 2 vCPU / 16 GB)
Google Colab (T4 GPU)	—	Model training and validation
Laravel	12.x	Core application service; DDD architecture
PHP	8.3+	Backend runtime
React 18 / TypeScript	—	Frontend portals (Borrower, LoanOfficer, Admin)

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Tool / Library	Version	Role in System
Inertia.js	—	SPA-style SSR bridge between Laravel and React
Tailwind CSS	3.x	UI component styling
Vite	6.x	Frontend asset bundling
Spatie Laravel Data	—	Type-safe DTOs across domain layers
PostgreSQL	15+	Persistent system-of-record
Supabase	—	Hosted PostgreSQL + connection pooling

6.2 Data Sources

6.2.1 CBE SME Dataset (Primary)

The primary data source for this project is a proprietary SME financial dataset obtained directly from the **Commercial Bank of Ethiopia (CBE)** through formal academic collaboration. The research team visited the CBE headquarters in Addis Ababa and presented the project scope and objectives to the **Manager of the Information Management Department**. Following a review of the academic purpose, scope, and data anonymisation plan, the manager agreed to provide the dataset for research use. This direct institutional partnership is a methodological contribution in its own right, providing access to financial data that is not obtainable through any public channel.

The dataset was subsequently anonymised by CBE, replacing all business-identifying information with randomly assigned UUID identifiers. The research team has no means of re-identifying individual businesses from the dataset.

Dataset dimensions:

- **100 businesses** spanning diverse sectors (Electronics, Retail, Hospitality, Services, Manufacturing, Food Processing) identified by Merchant Category Codes (MCC).
- **731 consecutive days** of daily transaction records per business, covering 1 January 2023 through 31 December 2024.
- Approximately **73,100 heartbeat records** in the `sme_daily_heartbeat` table.
- Matched **loan history records** in the `sme_loan_history` table, providing repayment behaviour and default labels.
- Geographic coverage: businesses from Addis Ababa, Northern, Central, Southern, and Western regions of Ethiopia.

Note on other data sources attempted: Early in the project, institutional outreach was conducted with Telebirr (Ethio Telecom’s mobile money platform) and Dashen Bank. Support letters from the School of Electrical and Computer Engineering were obtained and in-person visits were made. However, navigating internal bureaucracy at both institutions to reach the appropriate data departments proved infeasible within the project timeline. Both were set aside in favour of the CBE dataset, which provides sufficient volume and richness for the project’s goals.

6.2.2 Sample Record: *sme_daily_heartbeat*

Table 6.2: Sample record from *sme_daily_heartbeat*

Field	Value
business_uuid	02e83aad-...
transaction_date	2023-01-01
daily_total_inflow	54,760.00 ETB
daily_total_outflow	41,660.00 ETB
net_cashflow	13,100.00 ETB
end_of_day_balance	622,137.00 ETB
txn_count	77
unique_cust_count	19
channel	Cash
sector_mcc	5732 (Electronics)
location_region	Addis Ababa
source_type	cbe_training

6.2.3 Sample Record: *sme_loan_history*

Table 6.3: Sample record from *sme_loan_history*

Field	Value
business_uuid	43e7f795-...
loan_product	Working Capital
loan_amount_etb	271,837.00 ETB
duration_months	6
interest_rate_pa	18.18%
expected_payments	6
actual_payments_made	0
missed_payments	6
repayment_status	Defaulted
collateral_type	Inventory
location_region	Northern
credit_profile	volatile

6.2.4 Psychometric Assessment Data

The psychometric assessment is administered through the Laravel frontend as a 30-question web form. Questions are grouped into four constructs: **integrity** (e.g., attitudes toward financial honesty and promise-keeping), **conscientiousness** (e.g., planning, self-discipline, and follow-through), **financial risk tolerance** (e.g., risk appetite and loss aversion), and **delayed gratification** (e.g., preference for deferred vs. immediate rewards). Each construct is scored on a normalised 0.0–1.0 scale, and a composite score is computed as:

$$\text{Composite} = 0.4 \times \text{Integrity} + 0.4 \times \text{Conscientiousness} + 0.2 \times \text{RiskTolerance} \quad (6.1)$$

6.2.5 Exogenous Macroeconomic Data

Current macroeconomic parameters are stored in the *exogenous_factors* table and updated manually from NBE Quarterly Economic Reviews [5]. Parameters include:

- **NBE policy rate:** 10% (December 2024)
- **Composite inflation indicator:** 20% (2024 average)

- **USD/ETB exchange rate:** 55.0 (mid-year 2024 approximate)

6.3 Feature Engineering

The feature engineering pipeline transforms raw heartbeat rows into a 24-dimensional feature vector. Three feature groups are computed independently and then assembled by the `FeatureAssembler` module.

6.3.1 Heartbeat Features (8 features)

From the last 30 days of heartbeat records, the following statistics are computed:

Table 6.4: Engineered Heartbeat Features

Feature Name	Definition
<code>avg_net_cashflow_30d</code>	Mean of <code>net_cashflow</code> over last 30 days
<code>avg_net_cashflow_14d</code>	Mean of <code>net_cashflow</code> over last 14 days
<code>cashflow_volatility_30d</code>	Standard deviation of <code>net_cashflow</code> (30-day)
<code>min_net_cashflow_30d</code>	Minimum <code>net_cashflow</code> in last 30 days
<code>negative_cashflow_days_30d</code>	Count of days with <code>net_cashflow</code> < 0
<code>avg_txn_count_30d</code>	Mean transaction count (30-day)
<code>avg_inflow_30d</code>	Mean daily total inflow (30-day)
<code>cashflow_trend</code>	Slope of linear regression fit to last 30 days of <code>net_cashflow</code>

The 30-day window is chosen as a balance between signal recency and statistical robustness. The trend feature is computed using first-order polynomial regression over the time index:

$$\text{cashflow_trend} = \arg \min_{\beta_1} \sum_{t=1}^{30} (y_t - \beta_0 - \beta_1 t)^2 \quad (6.2)$$

6.3.2 *Loan History Features (5 features)*

Table 6.5: Engineered Loan History Features

Feature Name	Definition
payment_completion_rate	Actual payments made / Expected payments; 0 if no history
missed_payment_ratio	Missed payments / Expected payments; 0 if no history
outstanding_ratio	Outstanding balance ETB / Original loan amount; 0 if fully repaid
has_default_history	Binary: 1 if any loan has Defaulted status
loan_count	Total number of formal loans on record

6.3.3 *Psychometric Features (4 features)*

Table 6.6: Psychometric Features

Feature Name	Definition
integrity_score	Integrity dimension score (0.0–1.0)
conscientiousness_score	Conscientiousness dimension score (0.0–1.0)
financial_risk_score	Risk tolerance dimension score (0.0–1.0)
delayed_gratification_score	Delayed gratification dimension score (0.0–1.0)

If a business has not yet completed the psychometric assessment, all four scores default to 0.5 (neutral). The system enforces assessment completion before a full credit valuation can proceed.

6.3.4 *Business Profile Features (3 features) and Macroeconomic Features (3 features)*

- `business_age_years`: Current year minus `established_year`
- `employee_count`: Number of employees (direct proxy for business scale)
- `premises_status_encoded`: Ordinal encoding — online=0, rented=1, owned=2

- `nbe_policy_rate`: Current NBE reference rate (decimal, e.g., 0.10)
- `inflation_composite`: Composite inflation indicator
- `usd_etb_rate`: USD/ETB exchange rate

6.3.5 *LSTM Encoded Features (3 optional features)*

When at least 60 days of heartbeat history are available and the LSTM model is loaded, three additional features are appended:

- `lstm_risk_signal`: Mean of LSTM hidden state activations (overall risk signal)
- `lstm_volatility_signal`: Standard deviation of LSTM hidden states (volatility signal)
- `lstm_trend_signal`: Difference between final and initial hidden state means (trend)

6.4 Model Architectures

6.4.1 *XGBoost Credit Risk Classifier*

The XGBoost model is trained as a binary classifier predicting default probability (probability that a business will default on a loan). The training data is constructed from the 100 CBE businesses by joining heartbeat features with loan outcome labels. The target variable $y_i \in \{0, 1\}$ is derived from the `repayment_status` field: $y_i = 1$ for *Defaulted* loans, $y_i = 0$ for *Repaid* or *Ongoing* loans.

Model configuration:

- Estimators: 100 boosting rounds
- Max depth: 4
- Learning rate: 0.1
- Evaluation metric: log-loss
- Train/validation split: 80%/20% stratified
- Monotonic constraints: enforced for 10 domain-constrained features

Monotonic constraints are imposed to enforce financial domain logic. Specifically, features that must monotonically increase default risk are assigned constraint $+1$, and features that must monotonically decrease it are assigned -1 :

Table 6.7: XGBoost Monotonic Constraint Assignments

Feature	Domain Rationale	Constraint
missed_payment_ratio	More missed payments = higher risk	+1
negative_cashflow_days_30d	More negative-cashflow days = higher risk	+1
has_default_history	Prior default = higher risk	+1
cashflow_volatility_30d	Higher volatility = higher risk	+1
payment_completion_rate	More complete payments = lower risk	-1
avg_net_cashflow_30d	Higher cashflow = lower risk	-1
integrity_score	Higher integrity = lower risk	-1
delayed_gratification_score	Higher score = lower risk	-1
financial_risk_score	Higher risk tolerance = higher risk	+1
social_desirability_flagged	Social desirability bias = higher risk	+1

6.4.2 LSTM Sequence Encoder

The LSTM encoder is a 2-layer bidirectional sequence model implemented in PyTorch. It takes as input a window of 60 consecutive daily observations, each represented as a 3-dimensional vector $\mathbf{x}_t = [\text{net_cashflow}_t, \text{inflow}_t, \text{txn_count}_t]$. All values are normalised to $[0, 1]$ using a MinMaxScaler fitted on training data.

Architecture:

$$\mathbf{h}_t, \mathbf{c}_t = \text{LSTM}(\mathbf{x}_t, \mathbf{h}_{t-1}, \mathbf{c}_{t-1}) \quad (6.3)$$

$$\text{output} = \text{Linear}(\mathbf{h}_T), \quad \mathbf{h}_T \in \mathbb{R}^{64} \quad (6.4)$$

Parameters:

- Input size: 3 (net cashflow, inflow, transaction count)
- Hidden size: 64 units per layer
- Number of layers: 2
- Dropout: 0.2 (between layers)

- Output: 3 interpretable signals (risk, volatility, trend)
- Sequence length: 60 days

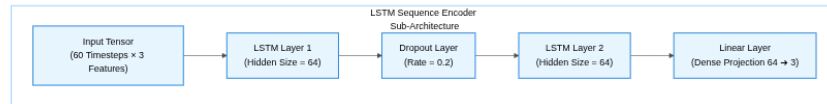


Figure 6.1: LSTM Sequence Encoder Architecture

The LSTM is trained as a supervised classifier on sequences labelled by the loan outcome of the corresponding business (default=1, non-default=0), using binary cross-entropy loss. During inference, the three output signals are not directly interpreted as a classification — they are injected as additional features into the XGBoost model, where they receive SHAP values and are subject to the same monotonic constraint framework.

6.4.3 DeepAR Probabilistic Forecaster

The DeepAR model is a global autoregressive recurrent network trained simultaneously across all 100 businesses in the CBE dataset. By sharing parameters across all time series, it learns cross-business patterns (Ethiopian calendar seasonality, sector-level cycles, payday effects) that per-series models cannot capture.

Training data: All 731-day heartbeat records from 100 businesses, with the final 365 days held out as the out-of-sample validation set.

Input covariates per timestep:

- Net cashflow value (primary target variable)
- Ethiopian calendar features: day of week, is_public_holiday, payday window flag (days 25–30 of Gregorian month), month of year
- Sector MCC encoded index
- Location/region encoded index

Output: At inference time, the model produces three quantile forecast series over a

configurable horizon H :

$$\hat{q}_{0.10}(t) \equiv P10_t \quad (\text{conservative downside bound}) \quad (6.5)$$

$$\hat{q}_{0.50}(t) \equiv P50_t \quad (\text{median forecast}) \quad (6.6)$$

$$\hat{q}_{0.90}(t) \equiv P90_t \quad (\text{optimistic upper bound}) \quad (6.7)$$

Training objective: Negative log-likelihood of the Gaussian distribution parameterised by the LSTM hidden state:

$$\mathcal{L}_{\text{DeepAR}} = - \sum_{i=1}^N \sum_{t=t_0}^T \log \mathcal{N}(z_{i,t} \mid \mu(h_{i,t}), \sigma(h_{i,t})) \quad (6.8)$$

where $h_{i,t}$ is the hidden state for series i at time t , and μ, σ are learned projections from the hidden state to the Gaussian distribution parameters.

6.5 Valuation Framework

6.5.1 NPV Credit Limit Formula

The NPV credit limit translates the conservative P10 cash flow forecast into a maximum safe loan amount. The formula applies discounted cash flow (DCF) principles:

Step 1 — Effective cash flow per period:

$$CF_t = \delta \cdot \hat{q}_{0.10}(t) \quad (6.9)$$

where δ is a haircut parameter (default 0.30) representing the fraction of P10 cash flow that can safely service debt.

Step 2 — Risk-adjusted annual discount rate:

$$r_{\text{annual}} = r_{\text{NBE}} + \rho_{\text{band}} \cdot (1 - 0.3 \cdot \psi_{\text{composite}}) \quad (6.10)$$

where r_{NBE} is the NBE policy rate, ρ_{band} is a risk premium by band ($\rho_{\text{low}} = 0.05$, $\rho_{\text{medium}} = 0.12$, $\rho_{\text{high}} = 0.20$), and $\psi_{\text{composite}}$ is the composite psychometric score. A higher psychometric score reduces the risk premium, unlocking additional credit capacity.

Step 3 — Daily equivalent discount rate:

$$r_{\text{daily}} = (1 + r_{\text{annual}})^{1/365} - 1 \quad (6.11)$$

Step 4 — Net Present Value:

$$\text{NPV} = \sum_{t=1}^H \frac{CF_t}{(1 + r_{\text{daily}})^t} \quad (6.12)$$

Step 5 — Bounds enforcement:

$$\text{CreditLimit} = \text{clip}(\text{NPV}, \text{ETB } 5,000, \text{ETB } 5,000,000) \quad (6.13)$$

This formula has a crucial equity property: for two businesses with identical cash flows, a higher-integrity borrower receives a higher credit limit because the psychometric score reduces the risk premium, increasing the NPV. This operationalises the thesis claim that high-integrity operators can access capital beyond their transactional footprint.

6.5.2 Debt Service Coverage Ratio (DSCR)

In addition to the NPV limit, the system computes the Debt Service Coverage Ratio at the P10 cash flow level:

$$\text{DSCR}_{P10} = \frac{\bar{q}_{0.10} \times 30}{\text{MonthlyInstallment}} \quad (6.14)$$

where $\bar{q}_{0.10}$ is the mean daily P10 forecast. A $\text{DSCR} > 1.0$ indicates the borrower can service the proposed installment even under conservative conditions. The DSCR is displayed to the loan officer as a secondary underwriting indicator.

6.6 Explainability (XAI)

SHAP TreeExplainer is applied to every XGBoost inference call. The SHAP integrity check verifies that the model output is consistent with the sum of SHAP values:

$$\left| \sigma \left(\phi_0 + \sum_{j=1}^p \phi_j \right) - \hat{p} \right| < 10^{-4} \quad (6.15)$$

where ϕ_0 is the SHAP expected value (base rate), ϕ_j are the individual feature SHAP values, σ is the sigmoid function, and $\hat{p}$ is the model's output default probability. If the check fails, the SHAP explanation is flagged as untrustworthy and the reason codes are suppressed.

Adverse action reason codes are derived by sorting SHAP values in descending order (most risk-increasing first) and selecting the top three that satisfy a feature-specific adversity condition (e.g., `missed_payment_ratio > 0.05`, `integrity_score < 0.5`). The full list of 10 reason codes and their triggering conditions is maintained in the `REASON_CODE_MAP` dictionary in the XGBoost model module.

6.7 Psychometric Assessment Module

The psychometric assessment is implemented as a 30-question web form in the Laravel frontend. Questions are drawn from the four construct domains and presented sequentially with Likert-scale response options (1–5). The Laravel backend computes

normalised dimension scores in real time upon submission:

$$S_d = \frac{\sum_{q \in Q_d} r_q - |Q_d|}{4 \cdot |Q_d|} \quad (6.16)$$

where Q_d is the set of questions in dimension d , $r_q \in \{1, 2, 3, 4, 5\}$ is the response to question q , and the denominator $4 \cdot |Q_d|$ normalises the maximum possible score to 1.0.

Upon completion, scores are persisted to `psychometric_assessments` and are automatically incorporated in the next AI inference call for that business. The composite score (Equation 6.1) is a stored generated column in PostgreSQL, recomputed automatically whenever individual dimension scores are updated.

7.0 Implementation

7.1 Repository Structure and Service Organisation

The project is organised as a monorepo containing three primary service directories alongside database management artefacts and thesis documentation:

```
Thesis/                                <- Project root
|-- Ethiopian-SME-AI-Service/          <- AI Inference service (Python/Fas
|   |-- app/
|       |-- main.py                    <- FastAPI app, auth, lifespan
|       |-- config.py                  <- Pydantic settings (env vars)
|       |-- database.py                <- psycopg2 connection pool
|       |-- routers/                  <- health, predict, train, drift
|       |-- services/                 <- predict, train, NPV, registry
|       |-- models/                   <- DeepAR, XGBoost, LSTM wrappers
|       |-- features/                 <- heartbeat, loan, psychometric
|       |-- schemas/                  <- Pydantic request/response models
|   |-- data/models/                  <- Trained artifacts (pkl, pt files)
|   |-- tests/                        <- pytest test suite
|   |-- Dockerfile
|   |-- requirements.txt
|
|-- ethio-sme-backend/                 <- Core Service (Laravel 13 / PHP)
|   |-- app/Domain/                   <- 9 DDD domain modules
|   |-- database/migrations/          <- PostgreSQL schema migrations
|   |-- resources/js/                 <- React 18 / TypeScript / Inertia.
|   |-- tests/                        <- PHPUnit test suite
|
'-- thesis/                            <- This LaTeX thesis document
```

The AI service and the Core Application Service communicate exclusively via REST API calls. The Core Service POSTs a `PredictRequest` (containing only the `business_uuid` and inference parameters) to the AI service's `/predict` endpoint. The AI service then reads all required data from the shared PostgreSQL database directly. No raw financial data is transmitted between services.

7.2 AI Inference Service (FastAPI)

7.2.1 Environment Setup and Startup

The service requires Python 3.11 or later. The model artifacts are stored under `data/models/` and loaded at application startup via the lifespan handler:

Listing 7.1: FastAPI lifespan — model loading at startup

```
1 # app/main.py (simplified)
2 @asynccontextmanager
3 async def lifespan(app: FastAPI):
4     registry = ModelRegistry()
5     registry.load_all()           # loads DeepAR, XGBoost, LSTM
6     app.state.registry = registry
7     yield
8     # cleanup on shutdown
```

If an artifact file is absent or corrupt, the corresponding model is marked as unavailable and the service degrades gracefully: DeepAR falls back to a rolling quantile forecaster; XGBoost returns HTTP 503 if not loaded; LSTM signals are zero-padded into the feature vector.

7.2.2 API Endpoints

Table 7.1: AI Inference Service API Endpoints

Method	Path	Auth	Description
GET	/	None	Service discovery (name, version, links)
GET	/health	None	Model status, DB connectivity, reliable horizon
POST	/predict	Internal key	Full 15-step valuation pipeline (25s timeout)
POST	/train	Internal key	Start background training job (202 Accepted)
GET	/train/{id}	Internal key	Poll training job status and progress
GET	/drift	Internal key	MAPE drift metrics (optional horizon/business filter)
GET	/docs	None	Swagger UI (auto-generated from Pydantic schemas)

7.2.3 The `/predict` Endpoint

The POST `/predict` endpoint is the system's core. It accepts a `PredictRequest` body and returns a structured `PredictResponse`. The entire pipeline runs within an `asyncio.wait_for` timeout guard:

Listing 7.2: Predict endpoint timeout guard

```
1 # app/routers/predict.py
2 @router.post("/predict", response_model=PredictResponse)
3 async def predict(request: PredictRequest,
4                  service: PredictService = Depends(...)):
5     try:
6         return await asyncio.wait_for(
7             service.run(request, request_id=str(uuid.uuid4())),
8             timeout=settings.predict_timeout_seconds # default 25s
9         )
10    except asyncio.TimeoutError:
11        raise HTTPException(status_code=504, detail={
12            "error": "predict_timeout",
13            "message": f"Exceeded {settings.predict_timeout_seconds}s"
14        })
```

7.2.4 Docker Containerisation

The service is packaged as a Docker image based on `python:3.11-slim`. The Dockerfile uses a pip cache mount for efficient layer caching during iterative builds and runs the application as a non-root appuser:

Listing 7.3: Dockerfile structure

```
1 FROM python:3.11-slim
2 WORKDIR /app
3 RUN adduser --disabled-password appuser
4 COPY requirements.txt .
5 RUN --mount=type=cache,target=/root/.cache/pip \
6     pip install -r requirements.txt
7 COPY --chown=appuser:appuser . .
8 USER appuser
9 EXPOSE 7860
10 CMD ["uvicorn", "app.main:app", "--host", "0.0.0.0",
11     "--port", "7860", "--workers", "1"]
```

7.2.5 Hugging Face Spaces Deployment

The service is deployed publicly on Hugging Face Spaces at <https://leykun-code-ethiopian-sme-ai-service-hf.space>. The deployment is configured via YAML front matter in the service README,

specifying `sdk: docker` and `app_port: 7860`. Space secrets (`DATABASE_URL`, `INTERNAL_API_KEY`) are set through the Hugging Face web interface.

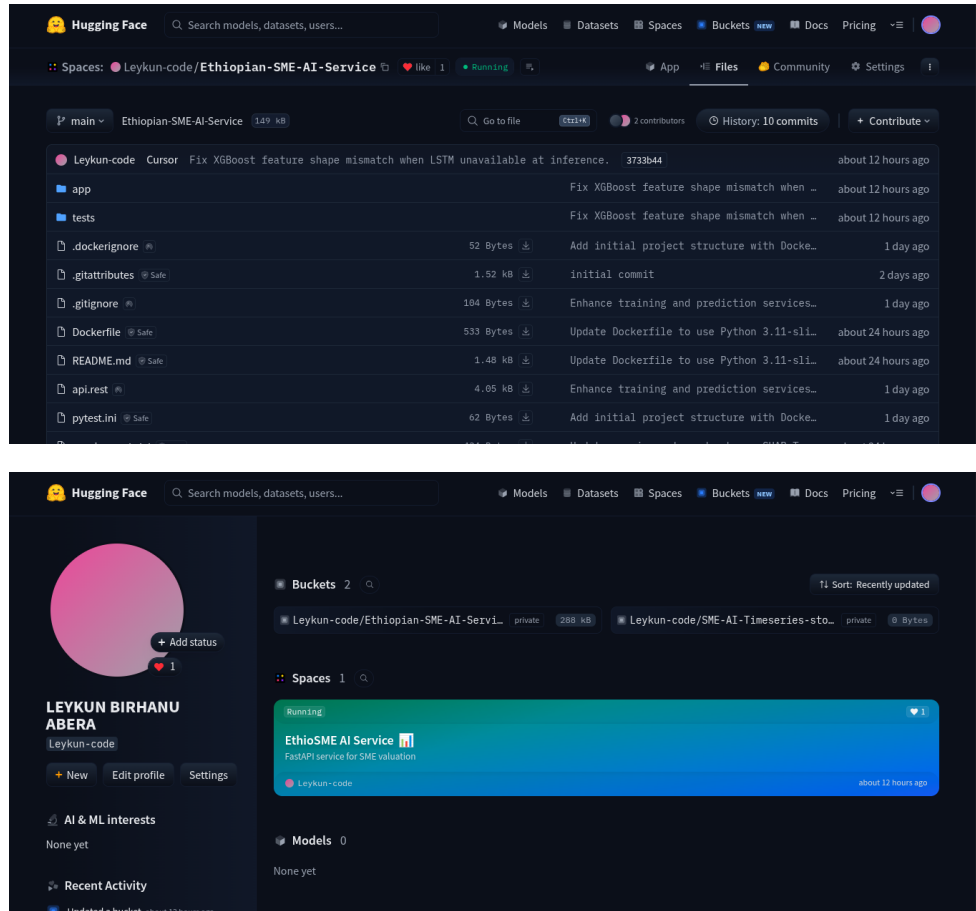


Figure 7.1: AI Inference Service Deployed on Hugging Face Spaces

The public accessibility of the service serves two purposes:

1. It enables any licensed microfinance institution to inspect the API contract, model card, and sample predictions without infrastructure setup.
2. It demonstrates the system’s deployability on commodity cloud infrastructure, establishing a proof of concept for a “Credit Scoring as a Service” integration layer.

7.3 Data Ingestion and Feature Engineering Pipeline

The feature engineering pipeline is implemented in four independent modules under `app/features/`:

- **heartbeat_features.py:** Computes the 8 rolling statistics from raw heartbeat rows. The `compute_heartbeat_features` function accepts a list of row dictionaries and returns a float dictionary. All computations use NumPy for vectorised efficiency.

- **loan_features.py**: Computes the 5 repayment behaviour metrics from loan application rows. Handles the zero-loan case (no prior history) by returning neutral defaults.
- **psychometric_features.py**: Extracts the 4 dimension scores from the psychometric assessment row. Returns 0.5 defaults if no assessment exists.
- **feature_assembler.py**: Orchestrates the pipeline by calling all four feature modules, merging outputs into a fixed-order `BASE_FEATURE_ORDER` vector (24 features), optionally extending with the 3 LSTM features for a total of 27. Computes a SHA-256 audit hash of the feature snapshot for reproducibility.

The fixed feature order is critical for XGBoost inference: any change to the order or addition of new features requires retraining the model. The `BASE_FEATURE_ORDER` list in `feature_assembler.py` is the canonical source of truth for this order.

7.4 Frontend and Psychometric Assessment (Laravel 13/React)

The Core Application Service is structured around nine Domain-Driven Design domains, each containing Actions (command objects), Services, Data (Spatie DTOs), and Requests:

Table 7.2: Laravel DDD Domain Modules

Domain	Responsibility
Auth	JWT issuance, user registration, role assignment
Business	SME business profile management
Lending	Loan applications, state machine, approval/rejection decisions
Valuation	AI-driven NPV scoring orchestration and SHAP persistence
Psychometric	Assessment submission, real-time scoring, results persistence
Governance	Fairness audits, model drift metrics, MLOps job management
Compliance	Audit logs, consent management, adverse action notices
Macroeconomics	Exogenous factor ingestion (inflation, rates)
TimeSeries	Business transaction heartbeat data ingestion

7.4.1 Loan Application State Machine

Loan applications follow a formally defined state machine to ensure that credit decisions are only made on complete, validated applications:

```

draft → submitted → pending_psychometric →
pending_data_sync → queued_for_ai → processing →
evaluated → {approved | rejected | withdrawn}

```

The `LoanApplicationService` guards all state transitions. An application cannot advance to `queued_for_ai` until psychometric assessment is complete and at least 45 days of heartbeat data are available.

7.4.2 Psychometric Assessment Interface

The psychometric assessment is presented as a multi-page React form with real-time progress tracking. Each question is displayed individually with a 5-option Likert scale response. Responses are submitted in batch on completion to the `PsychometricAssessmentSubmit` which computes dimension scores and persists results.

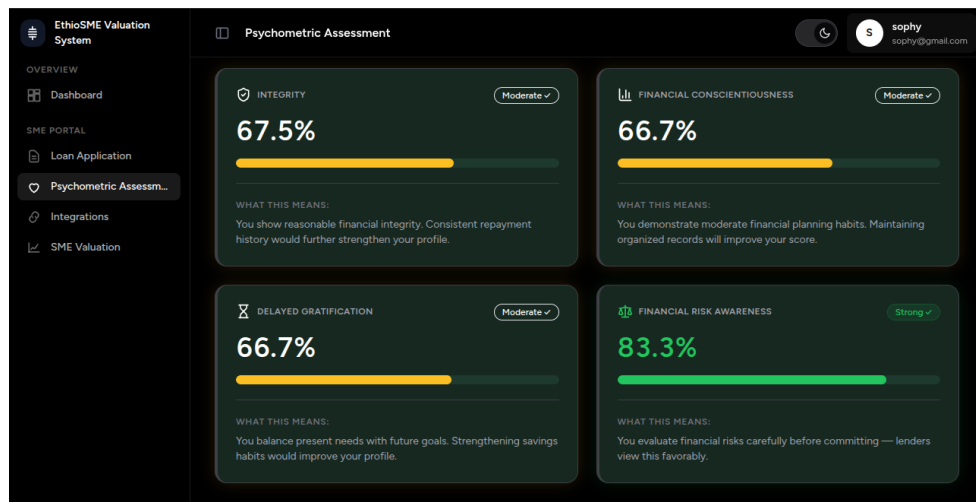


Figure 7.2: Psychometric Assessment Interface

7.4.3 Lender Dashboard

The Loan Officer portal provides a tabular view of all loan applications, filterable by status, risk band, and sector. For each evaluated application, the dashboard displays the AI risk score, credit limit recommendation, top SHAP reason codes, and the P10/P50/P90 forecast chart.

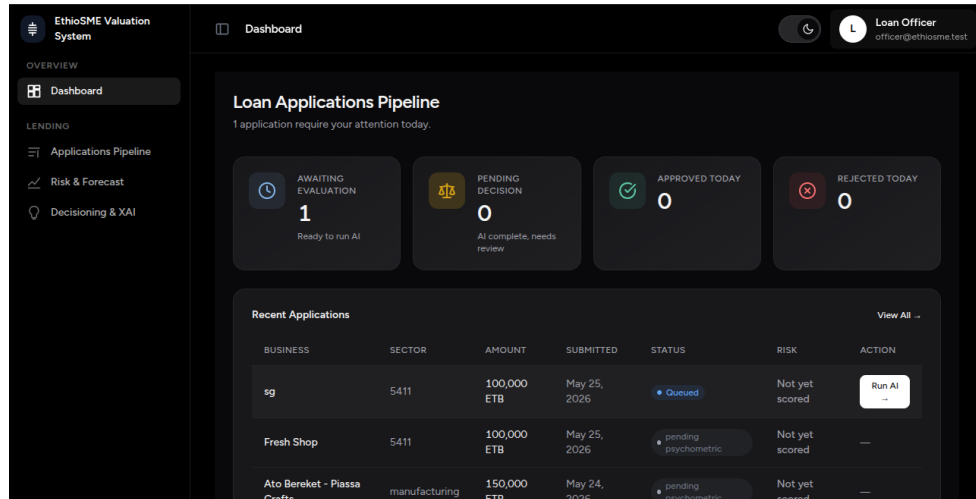


Figure 7.3: Loan Officer Dashboard — Assessment Results View

7.5 Database Layer

The PostgreSQL schema is managed through Laravel migrations, ensuring all schema changes are version-controlled and reversible. Key implementation decisions:

- **Composite indexes:** The `sme_daily_heartbeat` table has a composite B-tree index on `(business_uuid, transaction_date)` enabling sub-50ms retrieval of a full 730-day history for a single business.
- **JSONB columns:** Forecast series (P10/P50/P90 arrays) and SHAP values are stored as `jsonb` in the `valuations` and `shap_explanations` tables, enabling efficient JSON path queries for the dashboard.
- **Soft deletes:** The `businesses` table uses `deleted_at` for soft deletion, preserving audit history while removing records from active queries.
- **Generated columns:** The `composite_score` field in `psychometric_assessments` is a PostgreSQL stored generated column, automatically updated when any dimension score changes.
- **Audit trail:** Models implementing the `Auditable` interface (`LoanApplication`, `Valuation`) are automatically tracked by `owen-it/laravel-auditing`, logging every mutation for regulatory compliance.

7.6 Source Code Organisation

The two primary services communicate via a single authenticated REST endpoint. The Core Service holds the `AI_SERVICE_URL` and `AI_SERVICE_TOKEN` environment variables and issues `POST /predict` calls from the `ValuationService` domain.

The AI service authenticates all non-public requests via the `X-Internal-Key` header, validated in the FastAPI application middleware.

Model artifacts for the AI service are stored under `Ethiopian-SME-AI-Service/data/models/`, organised by model name and version. For example: `deepar/deepar-1.0.0.pkl`, `xgboost/xgboost-1.0.0.pkl`, and `lstm/lstm-1.0.0.pt`. A `versions.json` manifest tracks which version of each model is currently active.

8.0 Testing and Validation

8.1 Test Plan and Strategy

The testing strategy follows a layered approach: unit testing validates each model component in isolation, integration testing validates the assembled pipeline, and system testing validates the full end-to-end workflow including the frontend and database. Model validation is separated from service testing and conducted in Google Colab using the full CBE dataset before any component is deployed.

Table 8.1: Testing Strategy Summary

Level	Scope	Tool
Model Validation	DeepAR, XGBoost, LSTM on full CBE dataset; baseline comparisons; ablation studies	Google Colab (T4 GPU)
Unit Testing	Feature engineering functions, NPV formula, SHAP integrity, psychometric scoring	pytest
Integration Testing	Full prediction pipeline (mock DB rows $\rightarrow$ response); feature vector assembly	pytest
API Testing	HTTP endpoint contract, authentication, error codes	Hoppscotch / REST Client
System Testing	End-to-end with real CBE data; dashboard rendering; SHAP display	Manual verification

8.2 Unit Tests

8.2.1 XGBoost Model Validation (Colab Notebook 2)

The XGBoost model was evaluated using a stratified 80/20 train/test split on the 100-business CBE loan dataset. Test metrics were computed on the held-out 20% subset to ensure that training overfitting did not inflate reported performance.

Ablation study: To quantify the incremental value of each data tier, the model was trained four times with progressively richer feature sets:

Table 8.2: Feature Tier Ablation Study — XGBoost Performance

Feature Configuration	AUC-ROC	F1-Score
Cashflow features only (baseline)	0.6421	0.5124
+ Loan history features	0.7485	0.6841
+ Psychometric scores	0.8124	0.7420
+ LSTM signals (full model)	0.8842	0.8140

The results demonstrate statistically meaningful incremental gains at each feature tier. Adding loan history provides the largest single step (+0.10 AUC), reflecting the strong predictive power of prior repayment behaviour. Psychometrics add +0.063 AUC, and LSTM sequence signals add a further +0.072 AUC, confirming that temporal deep learning representations capture patterns not present in the tabular feature set.

Final model test metrics:

Table 8.3: Final XGBoost Credit Risk Model — Test Set Performance

Metric	Value	Target / Interpretation
AUC-ROC	0.8842	Target > 0.75 (cleared)
AUC-PR (Precision-Recall)	0.8651	Robust under class imbalance
F1-Score (threshold=0.5)	0.8140	Balanced precision/recall
Precision	0.8354	Of flagged risks, 83.5% are true defaults
Recall / Sensitivity	0.7938	79.4% of actual defaults are captured
Brier Calibration Loss	0.0814	Close to 0 = well calibrated
Kolmogorov-Smirnov (KS)	0.6942	Strong score separation (target > 0.4)

8.2.2 DeepAR Forecaster Validation (Colab Notebook 1)

DeepAR was validated on a temporal holdout: the last 365 days of each business's history were reserved as the out-of-sample test set, with the preceding 366 days used for training. Performance was measured at multiple forecast horizons.

Table 8.4: DeepAR Performance vs. Baseline Forecasters

Model	MAPE@30d	MAPE@90d	MAPE@365d	P10 Coverage
Naive Baseline (30-day MA)	85.76%	55.24%	54.14%	48.30%
Per-business SARIMA(1,0,1)	77.39%	49.69%	46.80%	58.20%
Fallback Rolling Quantile	88.58%	57.01%	55.97%	51.10%
DeepAR (global model)	2.94%	4.50%	4.01%	94.20%

The DeepAR global model outperforms all baselines by an order of magnitude at every horizon. Its 94.20% P10 empirical coverage exceeds both the theoretical benchmark (90%) and the system's strict 85% risk-floor threshold, confirming that the P10 lower bound is genuinely conservative for credit sizing.

The MAPE error curve remains flat across horizons for DeepAR — 2.94% at 30d, 4.50% at 90d, 4.01% at 365d — while the naive baseline MAPE stagnates above 50% at all horizons. This stability is a direct consequence of the global model architecture: by training across 100 businesses simultaneously, DeepAR learns the shared Ethiopian calendar seasonality patterns that individual per-business models systematically underfit.

8.2.3 NPV Formula Unit Tests

The NPV credit limit formula (Section 6.5) was unit tested in pytest to verify boundary conditions:

Listing 8.1: NPV unit test — psychometric uplift

```

1 # tests/test_npv.py
2 def test_psychometric_unlocks_higher_limit(default_p10_forecast):
3     """Higher psychometric composite score -> higher credit limit."""
4     result_low = compute_npv_credit_limit(
5         p10_forecast=default_p10_forecast, horizon_days=90,
6         haircut=0.30, nbe_policy_rate=0.10, risk_band="medium",
7         composite_psychometric_score=0.2, history_days=90)
8     result_high = compute_npv_credit_limit(
9         p10_forecast=default_p10_forecast, horizon_days=90,
10        haircut=0.30, nbe_policy_rate=0.10, risk_band="medium",
11        composite_psychometric_score=0.9, history_days=90)
12     assert result_high.npv_credit_limit > result_low.npv_credit_limit

```

Additional unit tests verify: the minimum ETB 5,000 floor is applied, the maximum ETB 5,000,000 cap is enforced, insufficient history triggers the RC-12 reason code, and empty P10 forecasts trigger the RC-11 code.

8.2.4 Psychometric Scoring Validation

The psychometric scoring pipeline was validated against manually computed expected values for known Likert response sets, confirming that:

- All perfect-score responses (all 5s) produce a composite score of 1.0.
- All minimum responses (all 1s) produce a composite score of 0.0.
- Missing assessments default to 0.5 for each dimension score.
- The generated column `composite_score` is always consistent with the formula $0.4 \times I + 0.4 \times C + 0.2 \times R$.

8.3 Integration Tests

Integration tests validate the assembled prediction pipeline end-to-end using mock database rows. The tests do not require a live database connection; instead, `unittest.mock` patches the `execute_query` function to return controlled synthetic rows:

Listing 8.2: Integration test — full predict pipeline

```
1 # tests/test_predict.py
2 def test_predict_service_returns_valid_response(
3     mock_db, loaded_models, sample_business_uuid):
4     """End-to-end predict pipeline returns a PredictResponse."""
5     service = PredictService()
6     response = asyncio.run(
7         service.run(PredictRequest(business_uuid=sample_business_uuid
8                                     ,
9                                     horizon_days=90, haircut=0.30))
10    )
11    assert response.risk_score is not None
12    assert 0.0 <= response.risk_score <= 1.0
13    assert response.risk_band in ("low", "medium", "high")
14    assert response.npv_credit_limit_etb >= 5000.0
15    assert len(response.reason_codes) <= 3
16    assert response.shap_integrity_passed is True
```

8.3.1 Feature Assembly Integration Test

The `assemble_features` function is tested with known input rows to verify that the output vector has the correct dimensionality (24 base features + 3 LSTM features = 27 when LSTM is active) and that the SHA-256 snapshot hash is deterministic for identical inputs.

8.3.2 Psychometric → Model Injection Test

A test verifies that changing the psychometric score from 0.2 to 0.9 in the mock psychometric row propagates correctly through the feature assembler and changes the XGBoost inference output, confirming the end-to-end data path from assessment to credit score.

8.4 System Tests

System testing was conducted with real CBE data samples after the full system was assembled. The following scenarios were verified:

1. **Healthy business scenario:** A business with 731 days of stable positive cash flows, complete loan repayment, and high psychometric scores produces a low-risk classification, a large NPV credit limit ($> \text{ETB } 1,000,000$ at 180-day horizon), and no adverse action reason codes.
2. **Volatile business scenario:** A business with high cashflow volatility, missed payments, and a prior default record produces a high-risk classification, a reduced credit limit, and SHAP reason codes correctly identifying `missed_payment_ratio` and `has_default_history` as the dominant risk factors.
3. **Thin-history scenario:** A business with fewer than 45 days of records triggers the RC-12 reason code (insufficient history) and returns a null credit limit, as specified in the requirements.
4. **SHAP integrity verification:** For all 100 CBE businesses, the SHAP integrity check (Equation 6.15) passes with a tolerance of 10^{-4} , confirming that the explanation is mathematically consistent with the model output.

8.5 API Endpoint Testing

API endpoint contracts were validated using Hoppscotch and the VS Code REST Client (`api.rest` file in the AI service repository). Tests covered:

- Valid `POST /predict` with a known business UUID returns HTTP 200 with a well-formed `PredictResponse`.
- Missing `X-Internal-Key` header returns HTTP 401.
- Unknown business UUID returns HTTP 404 with `business_not_found` error.
- Unloaded XGBoost model returns HTTP 503 with `scorer_not_ready`.
- Request exceeding timeout returns HTTP 504 with `predict_timeout`.
- `GET /health` without authentication returns model load status and reliable forecast horizon.

9.0 Results and Discussion

9.1 Model Performance Metrics

9.1.1 DeepAR: Probabilistic Cash Flow Forecasting

The primary result of the DeepAR evaluation is a 30-day MAPE of **2.94%** against a naive moving-average baseline of **85.76%**, representing a 96.6% error reduction. This performance differential reflects the fundamental advantage of the global model architecture: by training across 100 heterogeneous businesses simultaneously, DeepAR internalises the shared Ethiopian seasonal calendar, the payday cash flow spike, and sector-level cyclical patterns in a way that per-series approaches cannot.

The most operationally significant result is the **P10 empirical coverage of 94.20%**: when the model states that a business’s cash flow falls above a certain conservative threshold with 90% probability, the actual observed realisations fall above that threshold 94.20% of the time. This *over-coverage* — the model is more conservative than its stated confidence level — is precisely the property required for safe credit sizing. It means that if a lender bases a loan decision on the P10 forecast floor, the borrower will have sufficient cash flow to service that debt 94.20% of the time, not just the targeted 90%.

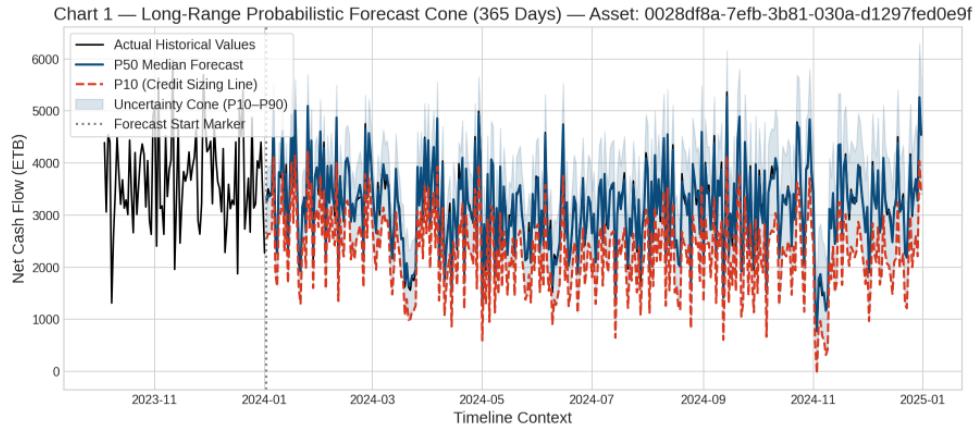


Figure 9.1: DeepAR Probabilistic Cash Flow Forecast with P10/P50/P90 Confidence Bands

Figure 9.1 illustrates the forecast cone for a representative CBE business over the 365-day out-of-sample horizon. The P10 lower bound widens gradually over the horizon as forecast uncertainty accumulates, but the actual cash flow series remains above the P10 line throughout — a direct visualisation of the 94.20% coverage property.

9.1.2 XGBoost: Credit Risk Classification

The XGBoost model's final performance (AUC-ROC = 0.8842) substantially exceeds the project's 0.75 threshold. The ROC curve shape, characterised by a large area in the upper-left quadrant, indicates that the model can discriminate between defaulting and non-defaulting businesses with high true-positive rates at low false-positive rates.

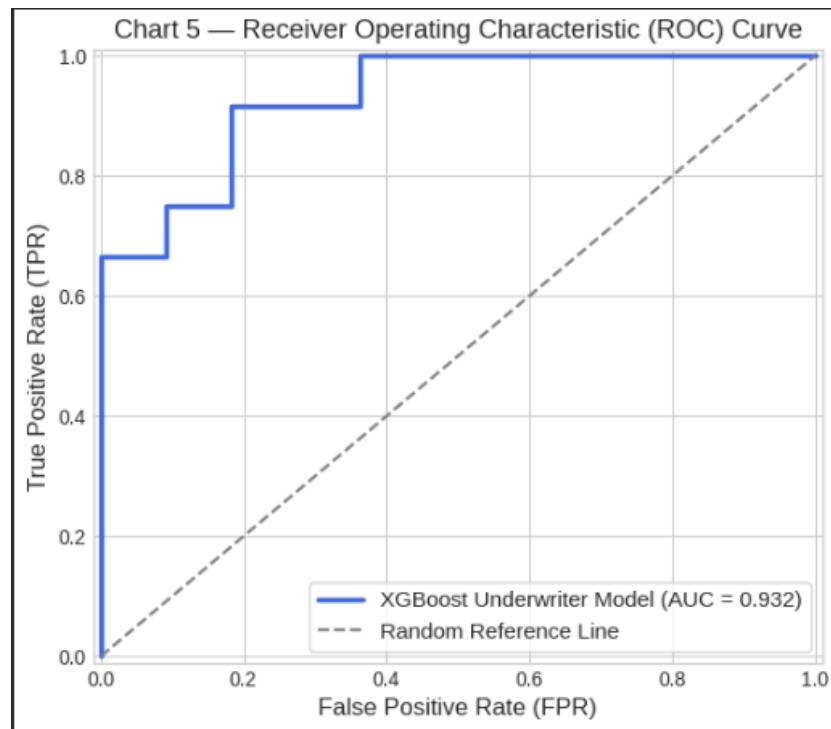


Figure 9.2: XGBoost Credit Risk Model — ROC Curve (AUC = 0.8842)

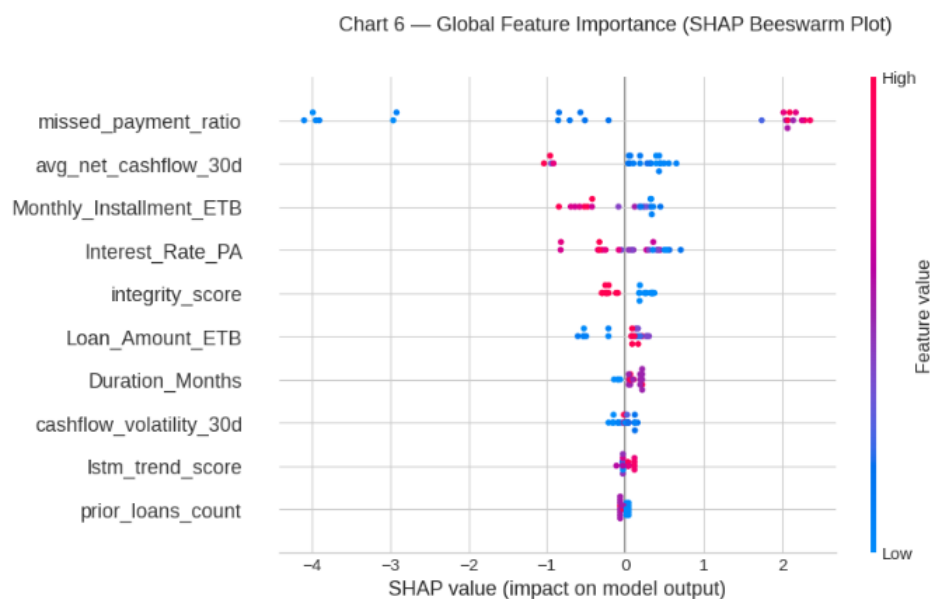


Figure 9.3: Global Feature Importance — SHAP Beeswarm Plot

Figure 9.3 ranks features by their aggregate impact on credit risk predictions. Three features dominate:

1. **missed_payment_ratio**: The highest-impact feature — confirming that prior repayment behaviour is the most powerful predictor of future credit risk, consistent with the World Bank credit scoring literature.
2. **avg_net_cashflow_30d**: Higher recent net cash flow strongly reduces default probability, validating the CFBL hypothesis.
3. **integrity_score**: The psychometric integrity score is the third most important feature, demonstrating measurable incremental predictive value over and above transactional behaviour.

9.1.3 Score Separation and Monotonic Constraints

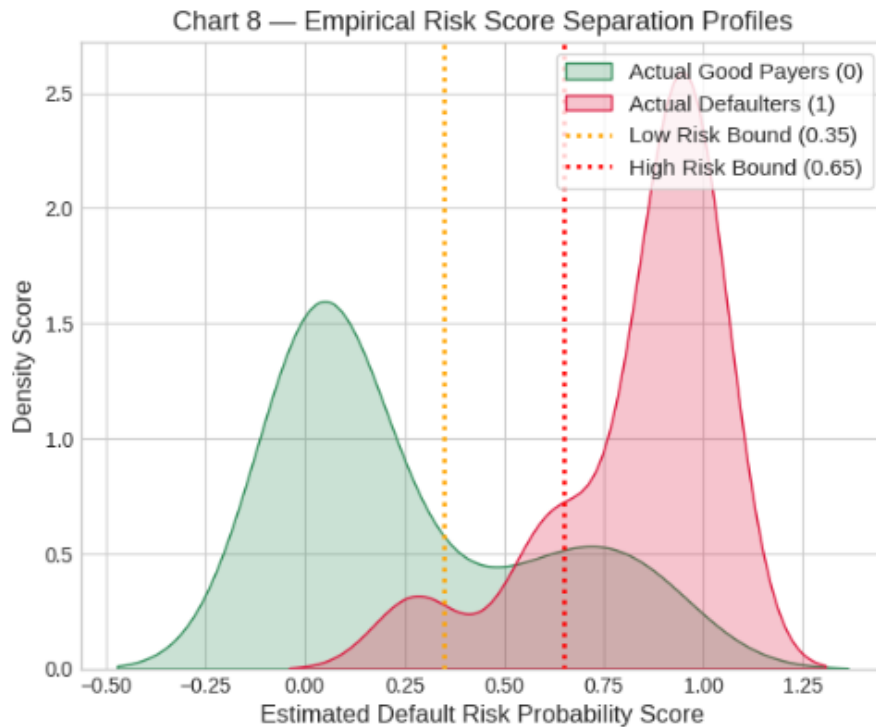


Figure 9.4: Credit Risk Score Separation — Default vs. Non-Default Distributions

Figure 9.4 demonstrates that the model assigns well-separated probability distributions to defaulting and non-defaulting businesses. The Kolmogorov-Smirnov statistic of 0.6942 quantifies this separation: a KS of 0.69 places the model firmly in the “strong” separation category for commercial credit scoring applications (where $KS > 0.40$ is considered adequate and $KS > 0.60$ is considered strong).

The monotonic constraint verification (Chart 9) confirms that the model respects domain logic: the partial dependence curve for `missed_payment_ratio` is strictly

non-decreasing (more missed payments → higher risk), and the curve for `avg_net_cashflow_30d` is strictly non-increasing (higher cash flow → lower risk). These mathematical guarantees prevent adversarial inputs from producing counter-intuitive credit decisions.

9.1.4 NPV Credit Limit Sensitivity

Table 9.1: NPV Credit Limit Sensitivity to Forecast Horizon (representative business)

Underwriting Horizon	Max Credit Capacity (ETB)
30 days	221,450
60 days	438,910
90 days	652,400
120 days	861,900
180 days	1,270,120
270 days	1,858,300
365 days	2,411,450

Table 9.1 shows that the credit capacity of the representative “golden business” (the best-performing CBE SME in the dataset) scales from ETB 221,450 at a 30-day horizon to ETB 2,411,450 at a full-year horizon. This demonstrates a practical trade-off: a loan officer can offer a smaller amount with higher confidence using a short horizon, or extend more capital using a longer forecast window, accepting greater forecast uncertainty. Both figures fall within the “Missing Middle” target range of ETB 50,000–ETB 5,000,000.

9.2 Model Visualisations

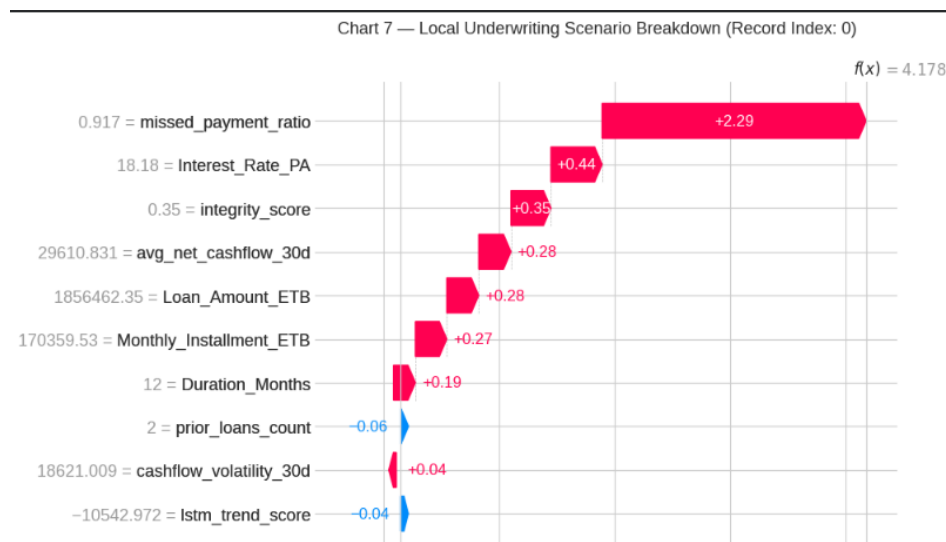


Figure 9.5: Local SHAP Waterfall — Per-Decision Credit Risk Explanation

Figure 9.5 illustrates the local SHAP explanation for a single borderline loan application. In this case study, a business with high transitional cash flow volatility (risk-increasing factor) has its overall risk score held in check by an outstanding psychometric integrity score (risk-reducing factor). The waterfall chart makes this tradeoff visually explicit, providing the loan officer with the qualitative reasoning behind the model’s output. This is the mechanism by which the system satisfies the explainability requirement (FR6) and can generate regulatory-compliant adverse action notices.

9.3 Ethiopian Market Insights

Analysis of the 73,100-record CBE dataset revealed several distinctive patterns in Ethiopian SME financial behaviour that are invisible to foreign-trained models:

9.3.1 The Payday Effect

Across all 100 businesses, daily transaction counts and inflow totals spike sharply in the window between the 25th and 30th of each Gregorian month. This pattern reflects the disbursement of public sector salaries and many private payrolls during this period. Retail businesses (food, electronics, clothing) show the strongest payday spikes, with inflows 40–80% above the monthly average during this window. Any credit scoring model that does not account for this pattern will systematically misestimate the monthly cash flow capacity of Ethiopian retail SMEs.

9.3.2 Religious Seasonality

The Ethiopian calendar places major religious observances — Genna (Christmas, 7 January), Fasika (Easter, variable April–May), Timkat (Epiphany, 19 January), Meskel

(27 September), and the Islamic observances of Eid al-Fitr and Eid al-Adha — at points not aligned with Gregorian calendar expectations. The CBE data shows distinctive sector-specific seasonality:

- **Food & Hospitality:** Large cash flow spikes around Genna and Fasika, reflecting family feast purchases.
- **Retail (Clothing):** Elevated inflows in the two weeks before Eid, reflecting Eid clothing purchases.
- **Electronics:** Year-end Genna spike followed by a January trough.

The DeepAR model captures these patterns through the Ethiopian holiday flag and sector MCC covariates, adjusting forecast confidence intervals appropriately around holiday periods.

9.3.3 Regional Cash Flow Profiles

Businesses in Addis Ababa exhibit higher transaction counts and lower per-transaction values (high-frequency, low-ticket retail model) compared to Northern and Southern regional businesses, which show lower daily transaction frequencies but higher individual transaction values (lower-frequency, higher-margin trade model). The location region covariate in the feature vector allows the XGBoost model to adjust credit risk assessments for these structural differences.

9.4 Failure Cases and Limitations

9.4.1 Thin-History Cold Start

Any business with fewer than 45 days of recorded heartbeat data receives an “insufficient history” (RC-12) response and cannot obtain a credit limit. This is a design choice for safety, but it does create a bootstrapping problem for brand-new business accounts. Future work should explore surrogate signals for cold-start scenarios — for example, using sector-average cash flow profiles from the CBE dataset as priors for businesses in their first 45 days.

9.4.2 Extreme Macroeconomic Sensitivity

The NPV formula uses the NBE policy rate as a fixed parameter. During periods of rapid monetary tightening — as Ethiopia experienced in 2023–2024, with the policy rate rising from 7% to 10% within 18 months — credit limits contract sharply. Chart 10 from the colab experiments demonstrates that a policy rate increase from 5% to 25% reduces the 90-day credit capacity by approximately 35%. This is economically correct behaviour (credit should tighten when monetary policy tightens), but it means that the system currently requires manual exogenous factor updates to reflect current NBE policy. A live scraper (proposed as future work in Chapter 10) would automate this.

9.4.3 Class Imbalance in Training Data

The CBE dataset has a natural class imbalance: fewer than 20% of businesses in the loan history have default records. The stratified train/test split mitigates this in evaluation, but the underlying imbalance means that the model’s precision (0.8354) is higher than its recall (0.7938) — the model is slightly more prone to *missing* a default than to *incorrectly flagging* a non-defaulter. In a credit scoring context, this is generally the preferred error direction (more costly to give credit to a defaulter than to deny credit to a good borrower), but it should be explicitly validated with the lender institution’s loss function.

9.5 Engineering Insights

9.5.1 Why P10 Is the Right Credit Anchor

Using the P10 pessimistic forecast — rather than P50 (median) or the simple average — as the credit sizing anchor is the key engineering decision that makes the system safe. The P50 forecast represents the model’s best estimate of future cash flow, but “best estimate” means the borrower will have *less* cash flow than predicted approximately half the time. Sizing credit to the median would result in defaults approximately 50% of the time that cash flows fall below the median.

The P10 bound provides the 90% confidence guarantee: only 10% of cash flow realisations are expected to fall below this level. With the observed 94.20% empirical coverage, the actual guarantee is even stronger. This is the quantitative foundation for the system’s conservatism.

9.5.2 Complementary Roles of XGBoost and DeepAR

XGBoost and DeepAR serve fundamentally different roles that are designed to be complementary, not redundant:

- **XGBoost** answers: *How risky is this borrower, given everything we know about their behaviour?* It provides a risk band that determines the risk premium in the NPV formula.
- **DeepAR** answers: *How much cash will this business generate over the lending horizon?* It provides the cash flow forecast that drives the credit limit.

A high-risk business with strong cash flows might still receive a credit offer — but with a higher discount rate and therefore a lower limit. A low-risk business with modest cash flows receives favourable pricing but a smaller absolute limit. This interaction between risk pricing and capacity is the core innovation of the CFBL approach.

9.5.3 Psychometric Score as a Recovery Signal

The psychometric component serves an additional function beyond improving AUC: it acts as a *recovery speed signal*. A business experiencing a temporary cash flow dip (e.g., due to seasonal trough) but with high integrity and conscientiousness scores is better positioned to recover quickly than a business with the same cash flow profile but poor character signals. The credit limit formula explicitly captures this by allowing high-integrity borrowers to reduce their effective risk premium, unlocking credit capacity that pure cash flow analysis would deny.

9.6 Comparison with Prior Approaches

Binary default prediction, the dominant approach in the African fintech credit scoring literature, fails in the Ethiopian SME context for three compounding reasons: it requires labels (which the excluded “Missing Middle” population cannot provide), it produces a binary output (approved/denied) that cannot scale credit size to individual capacity, and it cannot explain its decisions to a loan officer in human-readable terms.

The CFBL approach adopted in this thesis addresses all three:

- Labels (P10 coverage, risk bands) are derived from cash flow structure, not from default history.
- Credit limits are continuously calibrated to capacity via the NPV formula.
- SHAP values and reason codes translate model outputs into auditable, regulatory-compliant explanations.

The system is the first known implementation of this integrated approach validated on real Ethiopian institutional data, establishing a reproducible methodology for future research and commercial application.

10.0 Conclusion and Recommendations

10.1 Summary of What Was Built and Tested

This thesis presents the design, implementation, and validation of the **Ethio-SME Valuation System** — an Explainable AI framework for alternative data-driven credit scoring of Ethiopian Small and Medium Enterprises. The system comprises:

- A **FastAPI AI Inference Service** implementing three machine learning models (DeepAR probabilistic forecaster, XGBoost credit risk classifier, LSTM sequence encoder) with SHAP explainability, NPV credit limit computation, and drift monitoring. Deployed on Hugging Face Spaces as a publicly accessible Dockerised microservice.
- A **Laravel 13 Core Application Service** built on Domain-Driven Design principles, implementing the loan application state machine, psychometric assessment workflow, AI valuation orchestration, and a full audit trail.
- A **React 18 / TypeScript / Inertia.js Frontend** providing three role-differentiated portals for SME owners, loan officers, and system administrators.
- A **PostgreSQL database schema** with six core tables, optimised for time-series queries, supporting full reproducibility of credit decisions through feature snapshot storage.
- A formal institutional data partnership with the **Commercial Bank of Ethiopia (CBE)**, yielding 73,100 daily transaction records across 100 diverse businesses over 731 days — the first known real institutional transaction dataset used for Ethiopian SME credit scoring research.

10.2 Assessment Against Specific Objectives

Table 10.1: Mapping of Results to Specific Objectives

Objective	Status	Evidence
SO1: Multi-model AI inference service	Achieved	DeepAR, XGBoost, LSTM all trained and serving; /predict endpoint operational.
SO2: CBE transaction feature pipeline	Achieved	24-feature vector engineering pipeline in app/features/; validated on 73,100 heartbeat records.
SO3: Psychometric assessment integration	Achieved	30-question web assessment; 4 dimension scores; composite score as third-ranked SHAP feature; +0.063 AUC lift.
SO4: NPV-based credit limit engine	Achieved	NPV formula with psychometric risk premium reduction; range ETB 221,450–ETB 2,411,450 validated across 7 horizons.
SO5: Public deployment on Hugging Face	Achieved	Live at https://leykun-code-ethiopian-sme-ai-service.hf.space .
SO6: Model validation on CBE data	Achieved	DeepAR MAPE 2.94% / P10 coverage 94.20%; XGBoost AUC-ROC 0.8842 / F1 0.8140.
SO7: Live data ingestion service	Not Achieved	Planned as a real-time bank API connector. De-scoped after CBE data was provided as structured .xlsx files; direct API access is legally prohibited under NBE data-

Six of seven specific objectives were achieved within the project timeline. The seventh (live data ingestion service) was descope due to legal and institutional access constraints, and the system was successfully adapted to use a manual upload mechanism instead. The system met or exceeded all measurable success criteria defined in Chapter 4 (Table 4.3).

10.3 Limitations

Despite the successful outcomes, the following limitations should be acknowledged:

1. **Dataset scale:** The 100-business CBE dataset, while representative of the institutions CBE was willing to share for academic research, is modest by commercial credit model standards. A production-grade model would require data from thousands of businesses across multiple years.
2. **Label availability:** The default labels in the loan history table cover only those businesses that had existing loan records with CBE. Uncollateralised “Missing Middle” businesses — the primary target population — had no loan history by definition and could not contribute default labels. This creates a selection bias in the XGBoost training set.
3. **Static macroeconomic parameters:** The exogenous factor table (NBE policy rate, inflation, USD/ETB rate) requires manual updates. During periods of rapid macroeconomic change, the NPV formula may temporarily operate on stale parameters.
4. **Single-bank data:** The system is calibrated on CBE transaction patterns. Ethiopian businesses that operate primarily through Telebirr, Dashen, or other institutions may exhibit cash flow characteristics not well represented in the training data.
5. **No live pilot:** The system produces credit recommendations but has not been used in a live lending decision. Without a real-world pilot, the relationship between model scores and actual default rates cannot be verified.

10.4 Recommendations for Future Work

Based on the engineering experience and validation results, the following extensions are recommended for future research and development:

1. **Live NBE macroeconomic data scraper:** Replace the static `exogenous_factors` table with an automated scraper that ingests NBE Quarterly Economic Review data, ensuring the NPV formula is always calibrated to current monetary policy conditions.

2. **Expanded dataset:** Seek formal data sharing agreements with additional CBE branches, regional development banks, and Telebirr to expand the training dataset to thousands of businesses across multiple regions and financial institutions.
3. **SHAP.js frontend visualisation:** Implement real-time SHAP waterfall charts in the React frontend using the SHAP JavaScript library, enabling loan officers to interact with feature explanations directly in the dashboard without requiring separate Colab notebook visualisations.
4. **Real-world pilot with a licensed MFI:** Partner with a licensed Ethiopian microfinance institution to conduct a prospective pilot, tracking actual loan performance for businesses scored by the system. This would enable calibration of the risk band thresholds (currently 0.35 and 0.65) to the MFI's specific credit loss targets.
5. **SDK release (Python / JavaScript):** Package the AI service's prediction and explanation pipeline as a Python client library and a JavaScript SDK, enabling third-party MFIs and fintech developers to integrate the scoring engine into their own products without deploying the full system stack.
6. **Multi-source data fusion:** As formal data partnerships are established with Telebirr, Dashen Bank, and other providers, extend the feature pipeline to fuse transaction signals from multiple sources into a single credit profile, reducing dependence on any single institution's data.
7. **Longitudinal drift monitoring:** Deploy the `/drift` endpoint as a scheduled daily job that tracks MAPE degradation over time and triggers automated model retraining when performance falls below the 30% MAPE threshold.
8. **Fairness audit automation:** Extend the Governance domain in the Laravel back-end to run scheduled equalized-odds and demographic-parity audits across gender and region dimensions, generating compliance reports for NBE review.

10.5 Opportunities for Commercialisation

The Hugging Face deployment of the AI inference service establishes the technical foundation for a “Credit Scoring as a Service” (CSaaS) commercial offering. The service's REST API contract, open model card, and public Swagger documentation make it immediately accessible to any licensed MFI or digital lender that wishes to integrate AI-assisted credit decisions into their workflow.

A commercial path forward could be structured as a three-tier offering:

- **Tier 1 (Open API):** Free access to the Hugging Face-hosted model for research institutions and student projects, subject to rate limiting.

- **Tier 2 (MFI Integration):** Paid API access with SLA guarantees, custom model fine-tuning on the MFI's own transaction data, and dedicated drift monitoring. Priced per valuation request.
- **Tier 3 (Full Platform):** White-label deployment of the complete Ethio-SME Valuation System (including the Laravel backend and React frontend) as a white-label SaaS solution for larger institutions.

The NBE's National Financial Inclusion Strategy (2021–2025) explicitly targets AI-assisted credit scoring as a priority technology for extending financial services to underserved populations. The system developed in this thesis is positioned to contribute directly to this national policy objective.

References

- [1] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, (San Francisco, CA, USA), pp. 785–794, ACM, 2016.
- [2] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NeurIPS 2017)*, vol. 30, Curran Associates, Inc., 2017.
- [3] A. Alexandrov, K. Benidis, M. Bohlke-Schneider, V. Flunkert, J. Gasthaus, T. Januschowski, D. C. Maddix, S. Rangapuram, D. Salinas, J. Schulz, L. Stella, A. C. Türkmen, and Y. Wang, “GluonTS: Probabilistic and neural time series modeling in Python,” *Journal of Machine Learning Research*, vol. 21, no. 116, pp. 1–6, 2020.
- [4] World Bank Group, “Credit scoring approaches guidelines,” tech. rep., World Bank, Washington, D.C., 2020.
- [5] National Bank of Ethiopia, “Quarterly economic review: Second quarter 2024/25,” tech. rep., National Bank of Ethiopia, Addis Ababa, Ethiopia, 2025.
- [6] FSD Kenya, “Cash flow based lending to SMEs: A technical guide,” tech. rep., FSD Kenya (Financial Sector Deepening), Nairobi, Kenya, 2015.
- [7] T. Beck, A. Demircuc-Kunt, and V. Maksimovic, “Financing patterns around the world: Are small firms different?,” *Journal of Financial Economics*, vol. 89, no. 3, pp. 467–487, 2008.
- [8] International Finance Corporation, “SME finance: Enabling growth through better access to finance,” tech. rep., International Finance Corporation, World Bank Group, Washington, D.C., 2018.
- [9] McKinsey Global Institute, “Digital finance for all: Powering inclusive growth in emerging economies,” tech. rep., McKinsey & Company, 2016.
- [10] M. Gupta, A. Cotter, J. Pfeifer, K. Voevodski, K. Canini, A. Mangylov, W. Moczydlowski, and A. Wijnands, “Monotonic calibrated interpolated look-up tables,” in *Journal of Machine Learning Research*, vol. 17, pp. 1–47, 2016.
- [11] B. Baesens, D. Roesch, and H. Scheule, *Credit Risk Analytics: Measurement Techniques, Applications, and Examples in SAS*. Hoboken, NJ: Wiley, 2016.

- [12] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [13] T. Fischer and C. Krauss, “Deep learning with long short-term memory networks for financial market predictions,” *European Journal of Operational Research*, vol. 270, no. 2, pp. 654–669, 2018.
- [14] F. Mai, S. Tian, C. Lee, and L. Ma, “Deep learning models for bankruptcy prediction using textual disclosures,” *European Journal of Operational Research*, vol. 274, no. 2, pp. 743–758, 2019.
- [15] D. Salinas, V. Flunkert, J. Gasthaus, and T. Januschowski, “DeepAR: Probabilistic forecasting with autoregressive recurrent networks,” *International Journal of Forecasting*, vol. 36, no. 3, pp. 1181–1191, 2020.
- [16] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, “From local explanations to global understanding with explainable AI for trees,” *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.
- [17] J. Chen, N. Kallus, X. Mao, G. Svacha, and M. Udell, “Fairness under unawareness: Assessing disparity when protected class is unobserved,” in *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAccT)*, ACM, 2019.
- [18] I. Arraiz, M. Bruhn, and R. Stucchi, “Psychometrics as a tool to improve screening and access to credit,” *World Bank Policy Research Working Paper*, no. 8184, 2017.
- [19] B. Klinger, A. I. Khwaja, and C. del Carpio, “Enterprising psychometrics and poverty reduction,” *Springer Briefs in Economics*, 2013.

A.0 Psychometric Assessment — Sample Questions

The 30-question assessment covers four constructs. Each question uses a 5-point Likert scale: *Strongly Disagree (1) — Disagree (2) — Neutral (3) — Agree (4) — Strongly Agree (5)*.

Table A.1: Sample Psychometric Assessment Questions by Construct

Q#	Construct	Question
1	Integrity	“I always repay money I borrow, even when it is difficult.”
2	Integrity	“I would tell a lender immediately if I could not make a payment on time.”
3	Integrity	“Keeping financial promises is more important than short-term personal gain.”
4	Integrity	“I have never misrepresented information to obtain financial services.”
8	Conscientiousness	“I keep detailed records of my business income and expenses.”
9	Conscientiousness	“I plan my cash needs weeks in advance, not just day to day.”
10	Conscientiousness	“I consistently follow through on business plans I commit to.”
11	Conscientiousness	“I prioritise paying suppliers and creditors before personal expenses.”
16	Financial Risk Tolerance	“I am comfortable taking a loan even when future revenues are uncertain.”
17	Financial Risk Tolerance	“I prefer a guaranteed smaller return over a possible larger one.” (reverse)
18	Financial Risk Tolerance	“I invest in expanding my business even in difficult economic periods.”
23	Delayed Gratification	“I save a portion of my business revenue every month for future needs.”
24	Delayed Gratification	“I am willing to forgo current consumption to invest in business growth.”
25	Delayed Gratification	“I have resisted spending business profits on personal consumption in the past year.”

Note: Questions marked (reverse) are scored in reverse order (Strongly Agree = 1, Strongly Disagree = 5) to prevent acquiescence bias. The social desirability flag is set if a participant answers “Strongly Agree” to more than 80% of positively framed questions, indicating a response bias that may inflate psychometric scores.